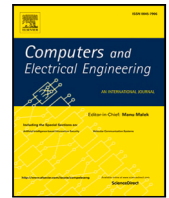




Contents lists available at ScienceDirect

Computers and Electrical Engineering

journal homepage: www.elsevier.com/locate/compeleceng

Machine learning-inspired intrusion detection system for IoT: Security issues and future challenges

Tariq Ahamed Ahanger^a, Imdad Ullah^b, Shabbab Ali Algamdi^c, Usman Tariq^a^a Department of Management Information Systems, CoBA, Prince Sattam Bin Abdulaziz University, Al-Kharj, Saudi Arabia^b School of Computer Science, Faculty of Engineering, The University of Sydney, Sydney NSW, 2006, Australia^c Department of Software Engineering, CCES, Prince Sattam bin Abdulaziz University, Al Kharj, Saudi Arabia

ARTICLE INFO

Keywords:

Internet of Things
security
Intrusion detection system
Deep Learning
Machine Learning

ABSTRACT

The Internet of Things (IoT) has revolutionized numerous domains, including smart grids, smart cities, healthcare, and business networks, by seamlessly integrating digital and physical systems. However, the rapid proliferation of IoT devices has introduced significant security challenges due to their resource constraints, heterogeneous architectures, and decentralized nature. Traditional security mechanisms, such as firewalls and IDS, often fail to address the unique vulnerabilities of IoT environments. This study provides a comprehensive analysis of the IDS market for IoT devices from 2014 to 2023, focusing on the evolution of deployment methods, emerging trends, and the integration of Artificial Intelligence (AI) strategies to enhance IoT security. The motivation for this research lies in the increasing reliance on IoT systems in critical infrastructures and the corresponding rise in sophisticated cyberattacks. Security breaches in IoT can lead to severe consequences, including data theft, service disruptions, and physical harm. To address these challenges, this study explores AI-driven techniques, such as Machine Learning (ML), Deep Learning (DL), and Federated Learning (FL), for detecting and mitigating complex intrusion patterns in IoT systems. By leveraging bibliographic analysis using VOS viewer, the study identifies key research themes, including blockchain-based security, DDoS mitigation, and cybersecurity for IoT, through keyword co-occurrence analysis with varying levels of overlap (50 to 250 keywords). This research also evaluates various IDS deployment methods, including Host-Based IDS (HIDS), Network-Based IDS (NIDS), and Hybrid IDS, based on metrics such as detection accuracy, resource efficiency, and adaptability to IoT environments. A detailed examination of IoT-specific intrusions, such as Sybil attacks, malicious node attacks, and memory exhaustion (DoS) attacks, is conducted to highlight vulnerabilities and propose AI-enhanced solutions. The novelty of this study lies in its integration of AI strategies into IDS frameworks, comprehensive market analysis over a decade, and systematic evaluation of IDS deployment tailored to IoT systems. The findings reveal that AI-driven IDS can significantly improve intrusion detection capabilities while addressing the resource constraints of IoT devices. This research provides actionable insights for researchers and practitioners, paving the way for the development of more robust and adaptive IDS frameworks to secure the rapidly expanding IoT ecosystem.

1. Introduction

The rapid proliferation of smart devices has been fueled by advancements in sensor technologies, automated item identification and tracking, device-to-device communication, and integrated Internet systems [1–3]. These innovations have enabled the

* Corresponding author.

E-mail addresses: t.ahanger@psau.edu.sa (T.A. Ahanger), imdad.ullah@sydney.edu.au (I. Ullah), s.algamdi@psau.edu.sa (S.A. Algamdi), u.tariq@psau.edu.sa (U. Tariq).

<https://doi.org/10.1016/j.compeleceng.2025.110265>

Received 15 January 2025; Received in revised form 9 February 2025; Accepted 8 March 2025

Available online 25 March 2025

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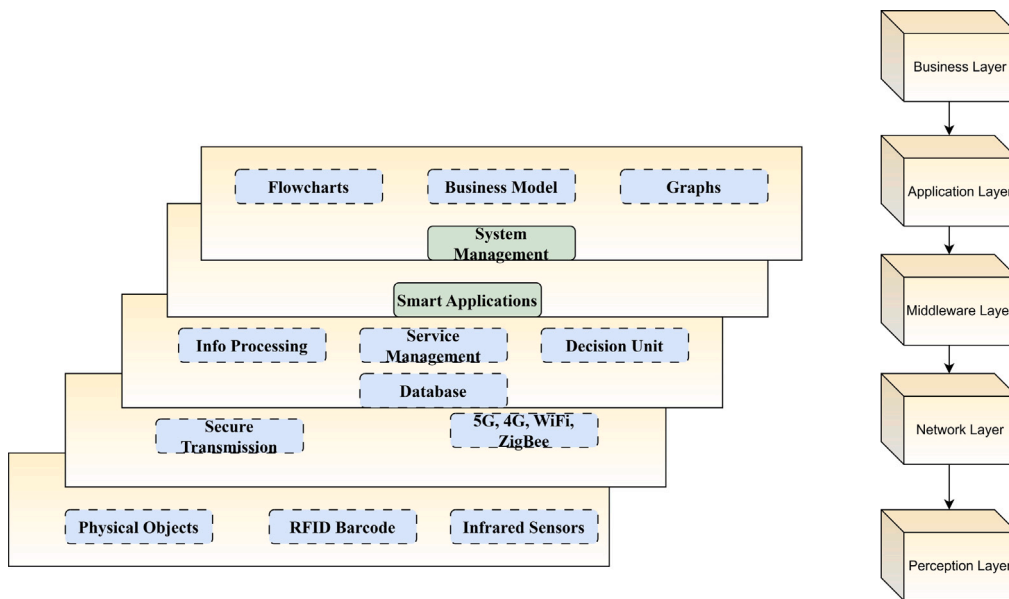


Fig. 1. IoT generic architecture.

development of Cyber-Physical Systems (CPS), with the Internet of Things (IoT) emerging as the interconnected network of these devices through the Internet [4–8]. According to Intel, IoT encompasses a vast array of intelligent sensors and devices that are transforming the web into a smarter, more interconnected ecosystem [8]. By 2033, it is predicted that the number of connected devices will reach 24 billion.¹ IoT has demonstrated immense potential across diverse domains, including industrial automation, logistics, healthcare, environmental monitoring, and safeguarding digital infrastructure [1,9,10]. However, the growing adoption of IoT devices has introduced significant security challenges for both networks and endpoints [10,11]. Security breaches in critical infrastructure, such as power grids and transportation systems, can disrupt daily life on a national scale. Similarly, false alarms in household appliances compromise individual privacy and security, leaving CPSs in essential infrastructure highly vulnerable to cyberattacks [12,13]. This research aims to address these security challenges by exploring robust solutions to safeguard IoT networks and their users from potential threats. Specifically, the study investigates the role of IDS in protecting IoT networks, identifying vulnerabilities, and proposing countermeasures to mitigate security risks. The vulnerabilities of IoT systems stem from the limited power and computational capabilities of connected devices. For example, performance assessment simulations involving devices like the Philips Hue light bulb and the Nest smoke alarm revealed that these constraints make IoT devices susceptible to breaches in security and privacy [14,15]. This underscores the need for robust security solutions that ensure users can benefit from IoT devices while adhering to stringent security standards [16,17]. Several techniques, such as firewalls, access control mechanisms, and privacy policies, have been implemented to secure IoT devices. These measures aim to protect data confidentiality, ensure validity, and restrict access to sensitive information [18–21]. Despite these efforts, IoT networks remain vulnerable to a wide range of cyberattacks. To address these vulnerabilities, robust security systems are required as the first line of defense. IDS plays a critical role in this context by monitoring device interactions within a network and raising alarms when suspicious activities or security flaws are detected [22,23]. As a vital component of IoT security, IDS provides a proactive defense mechanism to safeguard IoT networks against evolving cyber threats [22].

An Intrusion Detection System (IDS) is widely regarded as a critical security tool for conventional IP networks due to its ability to monitor traffic and raise alerts for suspicious activities [24–27]. While IDSs perform effectively in traditional networks, designing an IDS for an IoT network presents unique challenges [28–33]. These challenges arise from the inherent characteristics of IoT networks, such as limited data storage and processing power at individual IDS agent nodes [34–37]. IoT nodes are equipped with sensors, actuators, and communication interfaces, but lack the computational resources and flexibility typically assigned to IDS agents in conventional IP networks [38–43]. This makes it challenging to design IoT nodes with sufficient capabilities to ensure robust security. Unlike standard IP networks, where nodes have stable connections and reliably transmit data packets [19,20,41,44–47], IoT devices often operate with intermittent connections. To conserve power and bandwidth, IoT devices establish connections only at irregular intervals [48–51]. For instance, sensor nodes in IoT-based alert models typically exhibit both intermittent connectivity and limited transmission range [52–55]. Consequently, message relay within an IoT network relies on a series of intermediate nodes [56–60]. Data collected at a node must traverse multiple hops to reach its final destination within the network [31,61–66]. This complex architecture makes intrusion detection significantly more challenging, as IDSs must account for the dynamic and distributed nature

¹ Source: <https://www.statista.com/statistics/1194682/iot-connected-devices-vertically/>.

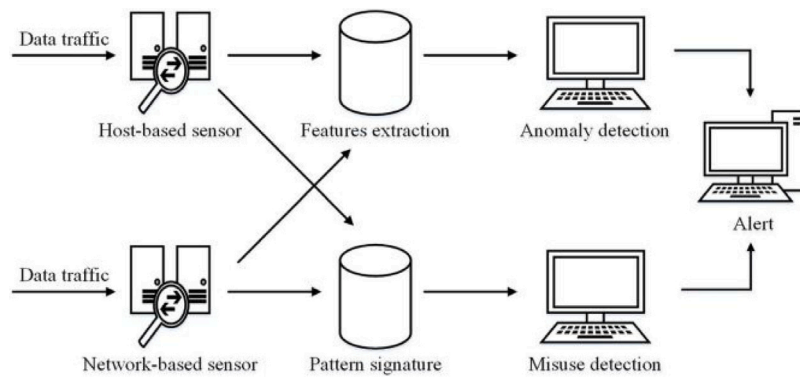


Fig. 2. IoT-IDS operational workflow: Conceptual perspective.

of IoT systems [67–72]. Furthermore, IoT networks operate across a wide range of protocols, each with distinct vulnerabilities and security requirements [73–75]. IoT-specific IDSs must be designed to address these diverse threats and mitigate their impact [76–80]. Effective IDS models for IoT should be capable of handling, categorizing, and correlating the multitude of alerts generated within the network to ensure efficient threat detection and response. To illustrate these concepts, a generic layered IoT architecture is presented in Fig. 1, while Fig. 2 provides an overview of the operational environment for IoT-specific IDSs.

1.1. Research motivation

IoT continues to evolve as a transformative technology with immense potential to revolutionize multiple domains. However, its rapid development also brings challenges, particularly in security and data privacy, which require careful attention to maximize its benefits while minimizing risks. Some of the motivational aspects include;

1. Wide Range of Applications:

- The IoT has become a prominent area of academic research due to its potential applications across a wide variety of fields.
- IoT is seen as a transformative technology capable of addressing numerous challenges faced by consumers, particularly in improving living conditions through modern technological advancements.

2. Technological Advancements Driving IoT Growth:

- The rapid adoption of cloud-based data management tools has significantly driven the growth of IoT.
- The ability to process and analyze massive amounts of data has made IoT technologies more accessible and practical for real-world applications.

3. Emergence of New IoT Applications:

- With the availability of affordable sensors, improved connectivity, and computational resources, IoT has enabled the development of various innovative applications.
- Some of the most promising IoT applications include:
 - **Smart Homes:** Automation of household tasks, energy efficiency, and enhanced security systems.
 - **Wearable Electronics:** Devices such as fitness trackers and smartwatches that monitor health and activity levels.
 - **Connected Automobiles:** Vehicles equipped with intelligent systems for navigation, safety, and autonomous driving.
 - **Industrial IoT (IIoT):** Enhancing productivity, automation, and predictive maintenance in industrial sectors.
 - **Agriculture:** Precision farming using IoT-enabled sensors for monitoring crops, soil, and weather conditions.
 - **Healthcare:** Remote patient monitoring, wearable health devices, and smarter medical equipment.
 - **Energy Management:** Efficient energy consumption and optimization through IoT-enabled devices.

4. Increased Focus on Security:

- As IoT applications expand, the need for robust security measures has become increasingly critical.
- The interconnected nature of IoT devices makes them vulnerable to cyber threats, necessitating advanced security protocols to safeguard users and their data.

5. Data Privacy and Ethical Concerns:

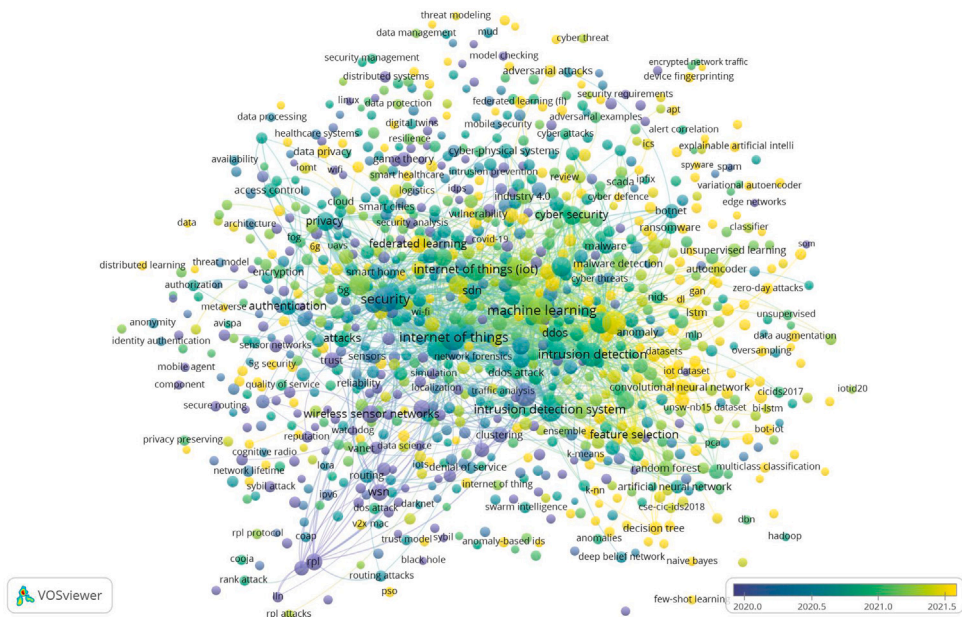


Fig. 3. Analysis of 50 overlapping keywords, highlighting shared themes, trends, and patterns across the datasets.

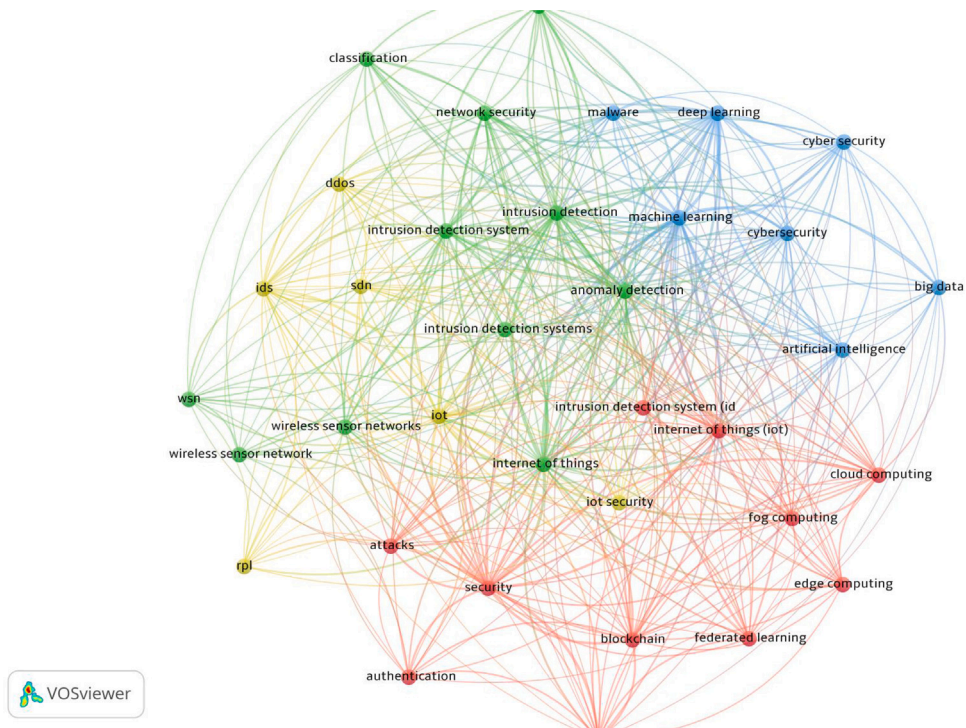


Fig. 4. Analysis of 100 overlapping keywords, highlighting shared themes, trends, and patterns across the datasets.

- IoT devices collect vast amounts of data, raising significant concerns about how this data is used, stored, and shared.
- Ensuring user privacy and establishing ethical guidelines for data usage are key challenges that must be addressed as IoT adoption continues to grow [81].

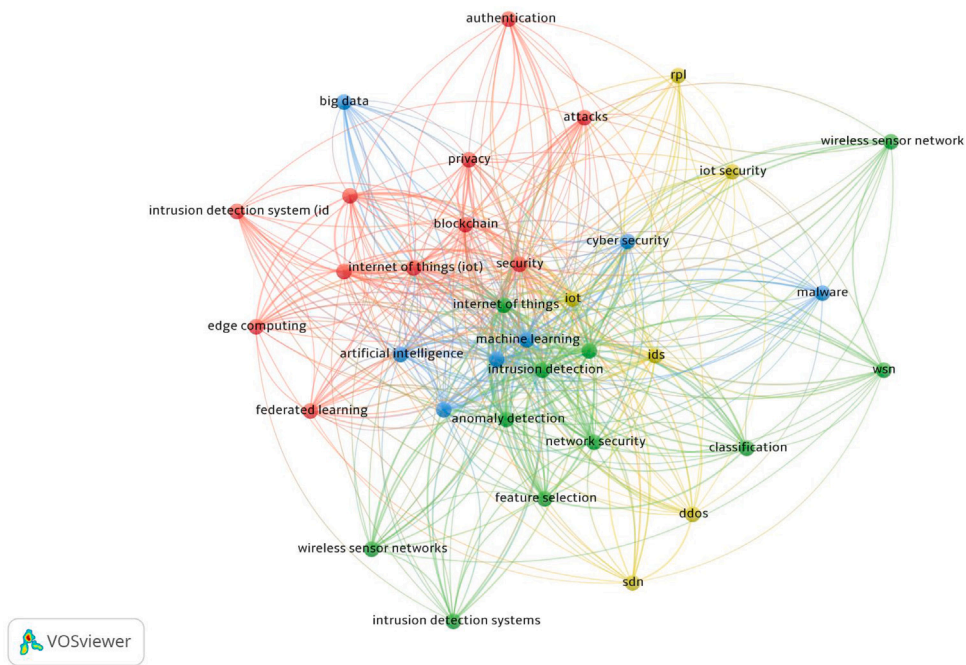


Fig. 5. Analysis of 150 overlapping keywords, highlighting shared themes, trends, and patterns across the datasets.

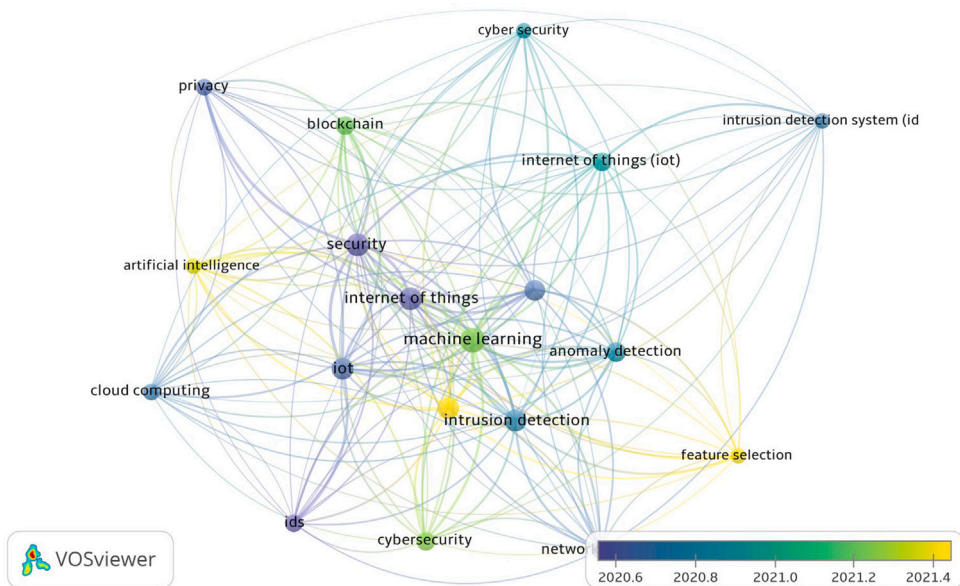


Fig. 6. Analysis of 200 overlapping keywords, highlighting shared themes, trends, and patterns across the datasets.

1.2. Major contribution

The primary contribution of this paper lies in addressing the critical need for more research on leveraging ML and DL techniques to enhance the security of IoT networks. The objective of this paper is to provide a comprehensive review of the existing work on IDS in IoT environments, with a particular focus on the application of ML and DL methods to construct IDS models. This research targets introductory-level researchers interested in IoT and its associated security concerns, offering insights derived from existing literature on intrusion detection in IoT networks. Furthermore, the paper identifies various types of intrusions that may occur in IoT networks and highlights the subsequent research challenges that have emerged. The scope of this research is defined as follows:

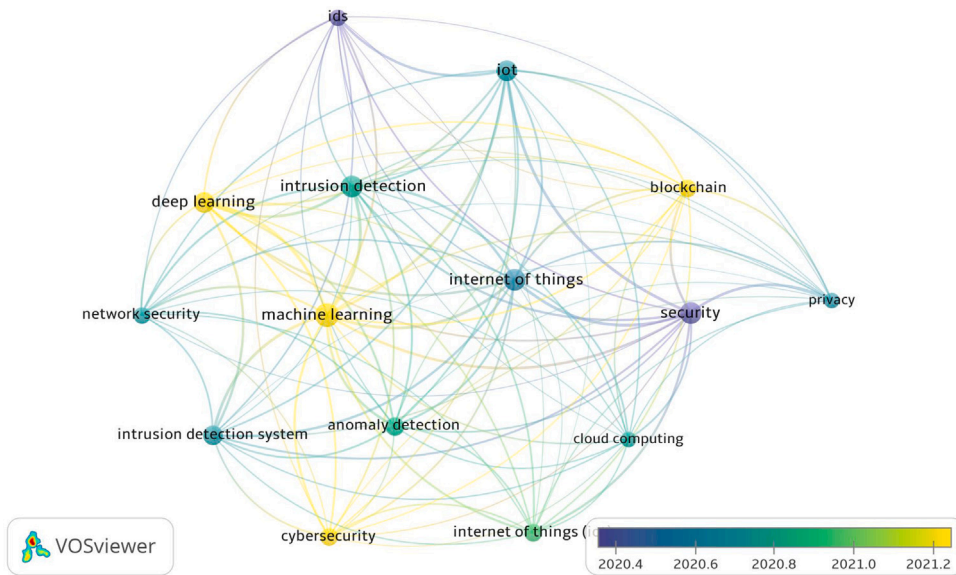


Fig. 7. Analysis of 250 overlapping keywords, highlighting shared themes, trends, and patterns across the datasets.

- To explore and analyze the existing literature on intrusion detection systems specifically designed for IoT environments.
- To investigate the application of ML and DL techniques for developing IDS models that can detect and mitigate security threats in IoT networks.
- To classify and categorize different types of intrusions in IoT networks and examine their impact on network security.
- To identify and discuss key research challenges and open issues in the domain of IoT security, particularly in the context of intrusion detection.
- To provide a taxonomy of IoT-based IDS and propose methodologies for analyzing data collected from IoT devices for intrusion detection purposes.

Based on the aforementioned objectives, the major contributions of this paper are as follows:

1. **Taxonomy of IoT-based IDS:** This paper presents a taxonomy of IDS for IoT networks. It discusses various approaches to deploying IDS in such networks and explains methodologies for analyzing data collected by IoT devices to detect potential intrusions.
2. **Categorization of Security Concerns:** The paper discusses IoT security concerns in terms of four broad categories: physical, network, software, and encryption. Each category is analyzed to highlight vulnerabilities and potential attack vectors.
3. **Application of ML and DL Approaches:** The study provides an overview of various ML and DL techniques that have been applied to develop IDS for protecting IoT networks. It evaluates their strengths, limitations, and applicability to different intrusion detection scenarios.
4. **Identification of Challenges and Research Gaps:** The research concludes by identifying key security concerns and obstacles associated with IDS in IoT networks. These include challenges related to scalability, resource constraints, adaptability to evolving threats, and the need for real-time intrusion detection.

1.2.1. Novel aspects

This study introduces several novel aspects that make significant contributions to the field of IoT security and IDS research. These include:

1. **Comprehensive Market Analysis (2014–2023):** A decade-long analysis of the IDS market for IoT devices, highlighting trends, advancements, and adoption across various IoT domains.
2. **Integration of AI Strategies:** Exploration of Artificial Intelligence (AI)-driven techniques, including ML, DL, and Federated Learning (FL), to enhance IDS frameworks specifically tailored for IoT environments.
3. **Keyword Co-occurrence Analysis Using VOS Viewer:** A bibliographic analysis that identifies key research themes and emerging trends in IDS for IoT, such as blockchain-based security, DDoS mitigation, and cybersecurity for IoT, using varying levels of keyword overlaps.
4. **Evaluation of IDS Deployment Methods:** A systematic evaluation of different IDS deployment methods, including Host-Based IDS (HIDS), Network-Based IDS (NIDS), and Hybrid IDS, based on detection accuracy, resource efficiency, and adaptability to IoT-specific constraints.

Table 1
Databases explored.

Database	Keywords	Records
IEEE explore	Anomaly intrusion detection AND Deep learning	114
IEEE explore	Anomaly intrusion detection system AND Deep learning	110
IEEE explore	Anomaly intrusion detection AND IoT	99
IEEE explore	Anomaly-based AND IoT	99
IEEE explore	Anomaly-based AND Deep learning	440
Science direct	Anomaly intrusion detection AND Deep learning	7
Science direct	Anomaly intrusion detection system AND Deep learning	5
Science direct	Anomaly intrusion detection AND Deep learning	2
Science direct	Anomaly intrusion detection AND IoT	2
Science direct	Anomaly intrusion detection system AND Deep learning	190
Scopus	Anomaly intrusion detection AND Deep learning	1252
Scopus	Anomaly intrusion detection system AND Deep learning	1345
Scopus	Anomaly intrusion detection AND IoT	1569
Scopus	Anomaly-based AND IoT	2515
Scopus	Anomaly-based AND Deep learning	3524
Web of science	Anomaly intrusion detection AND Deep learning	1100
Web of science	Anomaly intrusion detection system AND Deep learning	898
Web of science	Anomaly intrusion detection AND IoT	1356
Web of science	Anomaly-based AND IoT	1898
Web of science	Anomaly-based AND Deep learning	2565

- 5. Focus on IoT-Specific Intrusions:** Detailed analysis of IoT-specific security threats, such as Sybil attacks, malicious node attacks, and memory exhaustion (DoS) attacks, and how AI-enhanced IDS can effectively mitigate these vulnerabilities.
- 6. Addressing IoT Resource Constraints:** Examination of how AI-driven IDS frameworks can overcome the resource limitations of IoT devices while maintaining high detection accuracy and efficiency.
- 7. Integration of Emerging Technologies:** Discussion of emerging technologies, such as blockchain and federated learning, and their potential integration into IDS frameworks to address evolving IoT security challenges.
- 8. Practical and Actionable Insights:** The study provides actionable recommendations for researchers and practitioners to develop robust, scalable, and adaptive IDS frameworks for securing IoT ecosystems.

These aspects collectively make the study a significant contribution to the field of IoT security and IDS research.

This paper aims to bridge the gap between introductory-level researchers and advanced research on intrusion detection in IoT networks. By presenting a comprehensive review of IoT-based IDS, categorizing security concerns, and discussing ML and DL approaches, this paper provides a valuable resource for researchers and practitioners working in the domain of IoT security. Additionally, the identification of research challenges and open issues serves as a foundation for future studies aimed at enhancing the security of IoT ecosystems.

1.3. Research methodology

This study aims to analyze the current trends in IDS and their deployment methods, with a specific focus on IoT systems and AI-based strategies. To achieve this, a systematic methodology was employed for data acquisition, filtration, and evaluation. The following sections detail the methodology used for the market analysis and the evaluation of IDS deployment methods and AI strategies.

- 1. Data Acquisition** To gather relevant data for the study, a comprehensive search was conducted across multiple academic databases, including **Web of Science**, **Google Scholar**, **Scopus**, and **Science Direct**. Among these, **Scopus** was selected as the primary source for data acquisition due to its extensive coverage of peer-reviewed articles and high relevance to the research domain. A total of **9231 documents** were extracted from the last decade using carefully designed search strings related to IoT, IDS, and AI-based security strategies. These search strings included keywords such as “Intrusion Detection Systems”, “IoT Security”, “Machine Learning”, “Deep Learning”, and “Cybersecurity”. After applying selective filtration criteria, such as relevance to the research domain, publication quality, and recency, the dataset was refined to **355 articles**. This filtered dataset formed the foundation for bibliographic and keyword analysis.
- 2. Bibliographic Analysis and Keyword Co-occurrence** To identify the predominant research themes and trends in the domain, a bibliographic analysis was conducted using **keyword co-occurrence analysis**. This analysis was performed using the **VOS viewer**, a tool widely used for visualizing bibliometric networks. The following steps were undertaken:
- 3. Visualization Map Creation** Based on the identified keywords from the filtered dataset, a visualization map was generated to highlight the relationships and overlaps between key research terms.
- 4. Keyword Analysis** Keyword analysis is essential for identifying research trends, emerging themes, and gaps. By analyzing keywords like “IoT Security”, “Machine Learning”, and “Cybersecurity” through co-occurrence analysis and visualization maps, the study highlights dominant focus areas and evolving topics such as “Blockchain” and “Federated Learning”. This

process refines a large dataset (9231 documents to 355 articles) into relevant, high-quality research, ensuring the study's alignment with current trends. Tools like VOS Viewer provide insights into keyword relationships, enabling the study to contextualize its findings and contribute meaningfully to the field. The analysis was conducted with varying levels of keyword overlap (50, 100, 150, 200, and 250 overlapping keywords), providing progressively detailed views of the research focus areas:

- **Fig. 3:** Displays the keyword analysis with **50 overlapping keywords**, where the most relevant terms include “Internet of Things”, “Machine Learning”, “Security”, and “Intrusion Detection”.
- **Fig. 4:** Shows the keyword analysis with **100 overlapping keywords**, highlighting emerging themes such as “Blockchain” and “Federated Learning”.
- **Fig. 5:** Depicts the analysis with **150 overlapping keywords**, revealing a shift in focus toward “IDS”, “DDoS”, and “IoT Systems”.
- **Figs. 6 and 7:** Provide a broader view with **200 and 250 overlapping keywords** respectively, emphasizing research areas such as “Deep Learning”, “Machine Learning”, and “Cybersecurity”.

This analysis allowed the identification of key research trends and gaps, helping to contextualize the study's focus on IDS deployment and AI-based techniques.

5. **Evaluation of IDS Deployment Methods and AI Strategies** The study evaluated various IDS deployment methods and AI strategies to assess their effectiveness in IoT environments. The following steps were undertaken:
6. **Categorization of IDS Deployment Methods** IDS deployment methods were categorized into:

- **Host-Based IDS (HIDS):** Focused on individual devices.
- **Network-Based IDS (NIDS):** Monitored network traffic for anomalies.
- **Hybrid IDS:** Combined features of both HIDS and NIDS.

Each category was evaluated based on metrics such as detection accuracy, resource efficiency, and scalability in IoT systems.

7. **AI Techniques for IDS** A detailed review of AI techniques was conducted, focusing on:

- **ML:** Used for anomaly detection and classification.
- **DL:** Applied for complex pattern recognition in large datasets.
- **Federated Learning (FL):** Enabled decentralized training of models across IoT devices.

Emerging AI methods, such as **Reinforcement Learning** and **Blockchain-Integrated IDS**, were also explored for their potential to enhance security in IoT networks.

8. **Quantitative and Qualitative Metrics** The evaluation was based on:

- **Quantitative Metrics:** Detection rate, false positive rate, and computational efficiency.
- **Qualitative Metrics:** Ease of deployment and adaptability to IoT systems.

9. **Summary of Data Analysis** The findings from the bibliographic analysis and evaluation of IDS methods are summarized in [Table 2](#), which provides an overview of research explorations across different databases. The table highlights the number of articles reviewed, the predominant research themes, and the emerging trends in IDS and AI-based security strategies.

Algorithm for data analysis. The entire procedure for data acquisition, filtration, and analysis is outlined in **Algorithm 1**. This algorithm provides a step-by-step summary of the methodology, ensuring reproducibility and transparency in the research process. This methodology provides a structured approach to analyzing IDS deployment methods and AI strategies for IoT security. By leveraging bibliographic analysis, keyword co-occurrence, and a systematic evaluation of IDS techniques, the study identifies key trends and challenges in the domain. Future research can build on these findings to develop more efficient and adaptive IDS frameworks for IoT environments.

Paper organization. Section 2 depicts the related works in the current study domain. Section 3 provides a detailed taxonomy of IDS in IoT networks, including deployment strategies, analysis methodologies, and a concise overview of IoT. Section 4 examines various types of attacks that can occur within an IoT framework. Section 5 explores the application of ML and DL techniques for detecting attacks in IoT networks. Section 6 highlights the challenges and corresponding solutions in the development and implementation of IDS for IoT environments. Section 7 discusses potential future research directions in this domain. Finally, Section 8 concludes the study with a summary of key findings.

2. Related works

Several types of IoT technologies, including cloud computing [82–86], wireless sensor networks [87–90], and smart grid technologies [91–93], have been the subject of studies in the field of IDS. The issues with cloud computing security are briefly covered by Alam et al. [56]. In the article, the authors examined the security risks and concerns in a cloud computing environment by focusing on Platform as a Service and Software as a Service. In [94], the data life cycle in a cloud environment is outlined, and a quick assessment of security risks in the data life cycle is presented. Moreover, the authors have explored that security risks might be exploited in a cloud setting, and how this, in turn, can have repercussions for various actors in the cloud ecosystem, including

Algorithm 1 Research Methodology**Require:** Databases: Web of Science, Google Scholar, Scopus, Science Direct**Ensure:** Keyword Visualization Maps and Research Exploration

- 1: **Step 1: Data Acquisition**
- 2: Explore the following databases:
 - Web of Science
 - Google Scholar
 - Scopus
 - Science Direct
- 3: Select *Scopus* for data acquisition.
- 4: Extract **9231 documents** from the last decade using domain-specific search strings.
- 5: **Step 2: Data Filtration**
- 6: Apply selective filtration criteria to the extracted documents.
- 7: Identify **355 articles** for further analysis.
- 8: **Step 3: Bibliographic Analysis**
- 9: Perform **keyword co-occurrence analysis** on the filtered articles.
- 10: Use the **VOS Viewer** tool to create visualization maps based on the number of overlapping keywords.
- 11: **Step 4: Visualization Map Generation**
- 12: Generate visualization maps for different numbers of overlapping keywords:
 - **Figure 3:** 50 overlapping keywords.
 - Most relevant keywords: *Internet of Things, Machine Learning, Security, Intrusion Detection.*
 - **Figure 4:** 100 overlapping keywords.
 - Relevant keywords: *Blockchain, Federated Learning.*
 - **Figure 5:** 150 overlapping keywords.
 - Focus shifts to *IDS, DDoS, IoT systems.*
 - **Figure 6:** 200 overlapping keywords.
 - **Figure 7:** 250 overlapping keywords.
 - Predominant areas of research: *Deep Learning, Machine Learning, Cybersecurity.*
- 13: **Step 5: Research Exploration Summary**
- 14: Summarize research explorations for different databases in **Table 2**.

service providers and end users. For a comprehensive look at cloud-based intrusion detection methods and cloud-based attacks, several different kinds of cloud-based intrusions, as well as IDS detection methods and cloud-based IDS varieties, are discussed in Alharbi et al. [95]. The authors provide a brief overview of current cloud-based IDS, including a breakdown of their various forms, locations, data sources, and detection times. The survey details the benefits and drawbacks of each system, and it emphasizes the need to deploy IDS in the cloud network to deal with IDS security concerns. In Almomani et al. [69], a summary is presented of the several IDS methods that have been developed to detect intrusions in WSNs. There is a comparison of IDS types based on the detection measure, energy usage, computing required, and magnitude of false alarms created in the study.

Home and office networks can benefit from community-based IDS. This research classified IDS in terms of the type of detection method employed including fuzzy systems, neural networks, and support vector machines. The research in Brun et al. [96] is an in-depth look into the state of wireless sensor networks and the state of the art in terms of communication protocols and security concerns. Many smart grid applications for wireless sensor networks are outlined, as well as some of the difficulties those networks face in that context. The study provides a detailed comparison of the benefits and drawbacks of several different communication protocols used in wireless sensor networks, categorizing them by the protocol standard, spectrum type, and coverage range.

Several IDS strategies that have been deployed for remote networks are given in Buczak et al. [97]. The article explains the several types of availability, integrity, and confidentiality concerns that might affect wireless networks. The report details several attacks that have been uncovered, most of which are made possible by wireless networks' excessive need for power. The study also discusses other anomaly-based IDS strategies that have been applied in the literature, and it outlines the necessary components and architecture for building an IDS for wireless networks. In Chang et al. [86], an evaluation of six different smart grid systems is offered, with an emphasis on the need for a standard communication protocol to assure interoperability and security. The article provides an overview of smart grid applications and details the essential components of smart grid architecture. To achieve interoperability, the study stresses the importance of a standardized communication protocol. The report also describes the difficulties and key skill shortages that arise during the deployment of smart grid communication infrastructure. ICT technology including control, and

Table 2

Comparative analysis with state-of-the-art review works.

Ref	Major contribution	Survey domain	IDS-oriented	Research methodology	ML/DL	IoT	Data analytics
[100]	System test-bed analysis for security	Security	N	N	Y	Y	Y
[101]	ML based security technique analysis	Cybersecurity	N	NoT Specific	Y	N	Y
[102]	IDS system for IoT network	IDS	N	N	N	Y	Y
[103]	Cyber security techniques analysis	Cybersecurity	N	N	Y	Y	Y
[104]	IDS techniques	IDS	Y	N	Y	N	N
[105]	Security network exploration	Cybersecurity	N	N	N	Y	N
[106]	Anomaly identification with IoT Infrastructure	Cybersecurity in IoT	Y	N	Y	Y	N
[107]	IDS based ML techniques	IDS	Y	N	Y	Y	Y
This paper	ML/DL techniques for IDS in IoT network	ML for IDS	Y	Article Selection from Database	Y	Y	Y

energy usage are included in Aburomman et al. [92]'s overview of key difficulties in smart grid technology. ICT with automated control technology, and energy usage are discussed as they relate to the security concerns raised by the development of smart grid technologies throughout the years. Ciklabakkal et al. [98] does a quick analysis of cyber-physical smart grid testbeds. Developing a taxonomy and standards for attribute identification of smart grids is the focus of the research reported in Chawla et al. [99]. The study details a four-tiered taxonomy for smart grid technologies, research objectives, test environments, and communication networks. As well as discussing the current literature on smart grid testbeds, the study also discusses the obstacles and potential future research in smart grids. The article also assesses the strengths and weaknesses of existing testbeds regarding the ability to facilitate research, communicate, meet security needs, support protocols, and be accessed remotely.

2.1. Comparative analysis

A comparative analysis has been shown in Table 1. The referencing papers were acquired from the Scopus database that were most relevant to the current study domain. In most of the review articles, there is no specific research selection criterion. Moreover, it can be seen that very few articles have focused on the IDS aspect of the ML and DL-based IoT. Furthermore, many review papers have focused on the cyber security aspect of data protection. On the other hand, the presented paper has explored IDS-oriented articles that have incorporated ML/DL techniques for data protection.

3. Internet of things: Fundamentals

In the IoT, high-tech gadgets are physically integrated into interconnected networks [26,108,109]. The gadgets are networked together and automatically share extensive amounts of data. The convenience of utilizing everything from household to factory machinery is ensured by IoT networks [6,16,110]. Many of the features of the IoT systems are generic, while the systems themselves are tailored to specific uses [88,98,111]. Typically, there are three stages in an IoT design, and these include the gathering, sending, and processing of data [81,112,112]. At the collection stage, all the information gathered by the sensor nodes in the actual network is compiled [81,113]. The sensor nodes pick up on the short-ranged communication of the gadgets in the area. The recorded data is processed using communication protocols that have restricted data transfer speeds, storage capacity, and range. As a result, LLN [114,115] is a common name for transitional networks. In the transmission phase, the data obtained in the previous phase is sent to the intended recipients (such as software programs and end users) [4,28,116–121]. Ethernet, Wi-Fi, Hybrid Fiber Coaxial (HFC), and DSL are all used in the transmission of data [122–126,126].

A network that can carry data between far-flung devices is built using these technologies and Internet Protocol (IP) protocols [7,127–130]. Via gateways, the protocols used during the collecting phase are combined with the IP protocols used during the transmission phase [131–133]. During the processing phase, the data generated by the applications extract relevant information about the underlying physical network [134,134–136]. After this, steps are made to manage and control networked items [137–139]. In the processing stage, hardware and software are brought together and communicated with one another [129,140–142]. Standardized communication protocols for IoT devices have been the focus of research. The communication methods necessitate protecting gadgets and software [30,65,143,144,144,145]. IEEE 802.15.4, LoRaWAN, 6LoWPAN, Bluetooth Low Energy (BLE), Z-Wave, WirelessHART, RPL, CoAP, and Message Queue Telemetry Transport (MQTT) are some of the protocols that may be used [111,146,147]. Certain levels of the LLN protocol stack are addressed by the IEEE 802.15.4, 6LoWPAN, RPL, CoAP, and MQTT standards. Nevertheless, BLE, WirelessHART, Z-Wave, and LoRaWAN are only a few examples of IoT standards that describe vertically integrated systems.

To standardize the physical and MAC layers of low-speed LANs, the Institute of Electrical and Electronics Engineers (IEEE) came up with IEEE 802.15.4. A couple of the most talked about application protocols for IoT networks are CoAP and MQTT. A group called Constrained RESTful Environments (CoRE) created a set of protocols including Constrained Application Protocol (CoAP) to facilitate the movement of data across an LLN [148]. Similar to the HTTP protocol used in IP networks [149]. In LLN, communications of this type are handled via the CoAP protocol. CoAP's request-and-answer technique is used for bi-directional communication between devices and users [150]. Message Queuing Telemetry Transport (MQTT) [151] is a message transmission protocol that allows devices to exchange data via publish and subscribe actions. It is a lightweight protocol that consumes less power and sends and receives fewer data packets. As this protocol can be easily implemented in IoT networks, it is perfect for device-to-device communication [152]. The Bluetooth Special Interest Group [132,153–155] discovered BLE. It uses a variant of the Bluetooth standard optimized for low-power gadgets. BLE's layered architecture consists of the physical layer, which is responsible for the actual transmission and modulation of data bits, and the link layer, which links the various nodes in the network and makes the data available to the user [156–159]. Similar to a piconet, in which nodes take on the roles of master or slave and all slave nodes link to a single master node, BLE relies on a topology whereby nodes can also operate as bridges [160–162]. Logical Link Control and Adaptation Protocol (L2CAP) is what makes the link layer work [116,130,163]. The streamlined version of BLE uses this protocol [116,133,164]. It is responsible for multiplexing the information that has been sent down from higher tiers [32,165–167].

L2CAP's higher layers include the Generic Attribute Profile (GATT), which looks for relevant services, and the Generic Access Profile (GAP), which defines the actions that must be carried out on the devices [168]. WirelessHART is a wireless network protocol that was derived from the Highway Addressable Remote Transducer (HART) standard [32,169]. It is meant to be used for commanding manufacturing procedures using wireless networks [170,171]. It is a set of rules for how devices should talk to one another through radio waves [166]. It allows for the administration and control of processes and assets in wireless applications [67,172]. It is used as a standard for communication between industrial instruments and is compatible with the HART protocol [173,174]. This protocol employs a mesh topology in which all nodes are directly linked to one another. The goal of using a mesh topology for a network is to have backup paths available in case of a failure or a change in the environment [175].

Z-wave, a low-power protocol architecture, was designed for home automation [176–179]. It is a wireless networking standard designed specifically for use in connected homes. It enables the gadgets to communicate with one another and share data using instructions. Using a mesh architecture and message acknowledgment allows for bidirectional communication. The smart home hub is the central controller for many smart home gadgets. The only device with Internet access is the hub. Any home application may send a command to the hub, and the hub will relay that information to the appropriate networked device [180]. LoRa Alliance created LoRaWAN, a low-power wide-area network in which physical devices communicate with one another via a gateway [181]. The network server acts as the hub for all communications. Wireless connections connect physical devices to gateways, whereas central servers connect to the gateway via IP protocol. It is feasible for a physical device to connect with many gateways at once, and it is the job of the centralized server to process the communication packets and discard those that are unnecessary [182].

3.1. IoT-specific intrusion detection system

Several criteria, such as IDS deployment strategy, IDS analysis strategy, IoT intrusion kind, and IoT attack detection method, are used to classify IDS approaches into a taxonomy [108]. Fig. 8 shows the IoT-layered specific attacks.

3.1.1. IDS placement techniques

The three steps of data collection, transmission, and processing [183] form the foundation of IoT designs. Although the IoT architectures described in Jahanian et al. [184] differ in terms of specific features, they consistently utilize three levels: the perception layer, the network layer, and the application layer. The perception layer facilitates information sharing among the sensor nodes and devices within the Low-power and Lossy Networks (LLNs). Information gathered at the perception layer is transmitted to the application and end users through the network layer, which operates based on the transmission phase and communication protocols. Networking hardware, such as routers, integrates the protocols of the physical layer and the network layer. The application layer enables user interaction with the environment, leveraging the computing phase, which is central to augmented reality systems.

When designing Intrusion Detection Systems (IDS) for IoT networks, two primary deployment strategies are available. IDS nodes can be installed within routers to detect intrusions originating from the Internet and targeting devices within the network. However, this approach may lead to transmission overhead due to frequent alarms triggered by the IDS [185]. Alternatively, placing IDS nodes within subnets reduces network traffic but requires additional space for logging the resulting alarms [109]. To address such limitations, IDS nodes can be distributed across the network, increasing processing capacity while reducing the need for centralized oversight [186]. These approaches each have advantages and disadvantages when implementing IDS in IoT networks.

1. **Centralized IDS:** In a centralized IDS deployment strategy, IDS agents are located at a central device within the network, often a router. The central node maintains a log of all requests sent to other networking nodes and analyzes data exchanged between them. For instance, IDS agents installed within routers monitor all incoming and outgoing traffic [109,187]. However, this approach faces challenges in maintaining a single-node architecture and detecting threats across multiple nodes and devices within LLNs [188]. Consequently, there is a need for IDS strategies capable of monitoring data transfer rates at other low-capacity nodes in the network. A significant limitation of centralized IDS is vulnerability to attacks on the central node itself. For example, Lee et al. [189] proposed an IDS framework that examines data packets sent and received by connected devices to detect botnet attacks in IoT networks. Similarly, Lerman et al. [190] utilized a centralized IDS placement to protect against Denial-of-Service (DoS) attacks.

In these frameworks [191], the IDS analysis and response agents are hosted on a central device, while sensor nodes are distributed across LLNs. These sensors monitor network activity and send their findings to the central node. For instance, in Liu et al. [192], a centralized architecture was proposed where an IDS agent installed in the router detects physical network attacks. Instead of directly monitoring traffic, the framework employs a heartbeat protocol to ensure node availability and detect failures. While the heartbeat protocol increases network traffic, it requires minimal resources, such as memory and power, to function.

2. **Distributed IDS:** In a distributed IDS deployment strategy, IDS agents are installed on every IoT node. This requires nodes to be optimized for efficient intrusion detection. For example, McLaughlin et al. [193] proposed a lightweight distributed IDS framework that uses auxiliary shifting and early matching to reduce computational overhead during intrusion detection. The study demonstrated that the proposed algorithm outperformed the Wu–Manber (WM) algorithm, a widely recognized pattern-matching algorithm, in terms of speed.

Similarly, Muniyandi et al. [194] introduced a distributed IDS method that focuses on energy-efficient intrusion detection. This approach involves monitoring neighboring nodes using watchdog nodes, which are responsible for observing and logging network activity. For instance, in the case of sinkhole attacks on 6LoWPAN networks, the IDS nodes utilize trust and privacy mechanisms to detect malicious behavior. Nodes are arranged hierarchically, with some serving as leaders and others as followers. Alerts are raised in the event of an attack, and compromised nodes are isolated from the network.

3. **Hybrid IDS:** A hybrid IDS combines the advantages of centralized and distributed deployment strategies. In this approach, the network is divided into clusters, with each cluster containing a central IDS node, known as the cluster leader. The cluster leader monitors and manages the nodes within its cluster. For example, Muniyandi et al. [194] proposed a hierarchical clustering approach where cluster leaders oversee neighboring nodes' activities. Hybrid IDS systems require more resources than distributed IDS systems but select resilient nodes to serve as cluster leaders.

In Nweke et al. [195], a hybrid IDS strategy was introduced in which selected nodes act as IDS hosts to monitor local network traffic and detect intrusions. The IDS nodes operate based on predefined rules tailored to the behavior of each cluster. For instance, routers, that handle higher traffic volumes, follow specific rules distinct from those applied to other nodes. Similarly, Pajouh et al. [196] proposed a hybrid IDS that partitions the network into equal-sized clusters, with each cluster having a leader node to monitor traffic and detect intrusions.

Hybrid IDS systems, such as SVELTE [197], utilize centralized data analysis nodes and decentralized mini-firewalls to block malicious traffic. In this system, network nodes transmit RPL data to the central node for analysis, while also reporting malicious activity. Similarly, Rahman et al. [198] proposed a framework where nodes report observed changes in neighboring nodes to a centralized hub for intrusion detection. Routers and network nodes collaborate to analyze traffic and detect intrusions, with alerts sent to the central node when threats are identified. This approach reduces computational overhead while maintaining the ability to detect intrusions effectively.

3.1.2. Assessment strategies

The analytic approach used to identify intrusions categorizes an IDS into one of four subcategories: anomaly-based, signature-based, specification-based, and hybrid. IoT-specific intrusion detection strategies are discussed below.

1. **Anomaly-based IDS:** Anomaly-based intrusion detection systems identify attacks by detecting patterns of behavior that deviate from established norms. These systems analyze system activity to construct user profiles and flag any behavior that does not conform to the norm as suspicious. Anomaly-based IDS is effective in identifying zero-day attacks and attacks involving the misuse of system resources. To detect malicious actions, user behavior profiles are examined for deviations from typical behavior. However, false positives are a common issue in anomaly-based IDS [199]. User behavior profiles can be constructed using statistical or machine learning methods [115]. For instance, Santos et al. [200] proposed an anomaly-based IDS for use in a 6LoWPAN network to detect botnet attacks by monitoring TCP control fields, packet lengths, and the number of connections for each sensor node. Alerts are triggered when a node exceeds a calculated threshold. Selvi et al. [201] introduced computational intelligence methods to create user profiles for network devices, where unique profiles are generated for each device's IP address. This allows for precise analysis and detection of malicious activity. Similarly, Shi et al. [202] utilized energy consumption as a behavioral indicator for detecting malicious nodes in routing schemes. Nodes with abnormally high energy consumption are flagged as malicious and removed from the routing table. Sicari et al. [12] proposed an anomaly-based IDS for resource-constrained IoT networks, employing bit-matching techniques to extract features from data packets. Experiments demonstrated its effectiveness in detecting worm propagation, tunneling, SQL injection, and directory traversal attacks. Sharafaldin et al. [203] suggested a distributed anomaly-based IDS that compares network packet characteristics, such as size and data transfer rate, to detect attacks. Wormhole attacks in IoT networks can also be identified by analyzing system resources, such as the number of neighbors generated or control packets exchanged after an attack [202]. This approach achieved attack detection with 94% precision and identified attackers with 87% precision. The framework's low power and memory usage make it suitable for IoT networks.
2. **Signature-based IDS:** Signature-based IDS detects intrusions by comparing network traffic samples against a database of known attack signatures. If a match is found, an alert is triggered. These systems are effective against known threats and are relatively simple to implement. However, they cannot detect novel attacks or variations of existing attack signatures that are not present in the database [204]. Swankoski et al. [205] proposed a signature-based IDS based on the Artificial Immune System,

where immune cells identify intruders using the unique characteristics of attacks. The system adapts to changes in network infrastructure but faces challenges in maintaining attack signatures and metadata.

The EBBITS project [55] and Taivalsaari et al. [206] developed signature-based IDS for detecting DoS attacks in 6LoWPAN networks. Suricata IDS was used to generate warnings forwarded to a DoS projection manager, which analyzed factors such as channel interference and dropped packets to identify intrusions. Thakkar et al. [14] introduced a multiple-pattern detection method to reduce the computational burden of matching attack signatures. Experiments demonstrated superior performance compared to standard pattern-matching methods.

3. **Specification-based IDS:** Specification-based IDS profiles networking components, such as nodes, routers, and servers, based on predefined criteria, including protocol and routing table details. Any deviation from these predefined norms triggers an alert. Unlike anomaly-based IDS, the profiles in specification-based IDS are manually defined by security professionals [207]. This approach results in low false positive rates and does not require training to understand network behavior. However, the manual definition of specifications can lead to delays in responding to changes and potential errors [208]. Xie et al. [209] proposed a specification-based IDS for detecting DDoS attacks in IoT networks by setting maximum thresholds for middleware layers. Alerts are triggered when resource requests exceed the specified limits. Zhang et al. [210] used a finite-state machine to monitor network activity for signs of intrusion based on user-defined specifications. Zidi et al. [211] also employed a finite state machine to identify RPL attacks, achieving high detection accuracy and low false alarm rates with minimal energy consumption. The effectiveness of specification-based IDS depends on the expertise of the administrator defining the specifications, as poorly defined rules can lead to high false alarm rates [212,213].
4. **Hybrid IDS:** Hybrid IDS systems combine the strengths of signature-based, anomaly-based, and specification-based IDS to achieve high classification accuracy and detection rates. Chowdhury et al. [214] developed a hybrid IDS to detect RPL attacks while balancing the memory usage of signature-based IDS and the computational cost of anomaly-based IDS. Edmond et al. [215] demonstrated that combining both IDS methods into a single framework improves the ability to counter a wider range of threats. Sharma et al. [216] proposed a hybrid IDS combining anomaly-based and specification-based methods to analyze network traffic and calculate nodes' reputation and trustworthiness. Nodes with a trust score above 0.5 were deemed safe. The system achieved a detection accuracy of 92% in wired networks and 75% in wireless networks, with a low false alarm rate.

4. IoT network-based intrusion: Proposed taxonomy

Attackers can compromise IoT networks through various methods, such as exploiting the physical architecture, probing communication protocols for vulnerabilities, or injecting malicious code to bypass encryption safeguards. To address these threats comprehensively, it is essential to develop a systematic taxonomy of intrusions. **Figure 9** illustrates this taxonomy of IDS, categorizing intrusions in IoT networks into four distinct types based on their objectives: physical, network, software, and encryption-based intrusions [217].

Taxonomical Categorization: These categories were chosen to reflect the specific components of IoT systems that attackers typically target, offering a clear and actionable framework for identifying and mitigating threats. Unlike earlier taxonomies, which often relied on broader or generalized classifications, this taxonomy prioritizes the unique vulnerabilities inherent to IoT environments. Each category is defined as follows:

- **Physical Intrusions:** These target the hardware or physical interfaces of IoT devices, such as tampering with sensors or exploiting insecure physical access points.
- **Network Intrusions:** These involve attacks on communication protocols, such as intercepting or manipulating data during transmission.
- **Software Intrusions:** These focus on exploiting vulnerabilities in the software layer, such as injecting malicious code or exploiting unpatched applications.
- **Encryption-Based Intrusions:** These aim to bypass or weaken cryptographic safeguards, compromising data integrity or confidentiality.

Novel Aspects: This refined categorization provides a more precise understanding of intrusion methods compared to previous models. Earlier taxonomies often grouped threats into broader categories, which may not have fully accounted for the complexity and interconnectivity of IoT systems. By focusing on specific objectives and components, this taxonomy enables the design of tailored Intrusion Detection Systems that address each type of threat effectively.

This taxonomy highlights the need for IDS strategies that are not only comprehensive but also specific to the nature of the intrusion. By differentiating between physical, network, software, and encryption-based threats, this approach enhances the overall security posture of IoT networks, ensuring a more robust defense against evolving cyber threats.

Types of Intrusions and Their Impact

Physical intrusions. Physical attacks target IoT devices and infrastructure, often involving tampering with hardware or disrupting power supplies. For example, attackers may physically access sensors to alter their readings or disable them entirely. Such intrusions can halt ongoing operations and corrupt data stored in system databases. Solutions include:

- **Tamper-Resistant Hardware:** Devices equipped with tamper-proof enclosures and anti-tamper sensors (e.g., ARM TrustZone).
- **Physical Access Control:** Biometric authentication and surveillance systems to restrict unauthorized access.

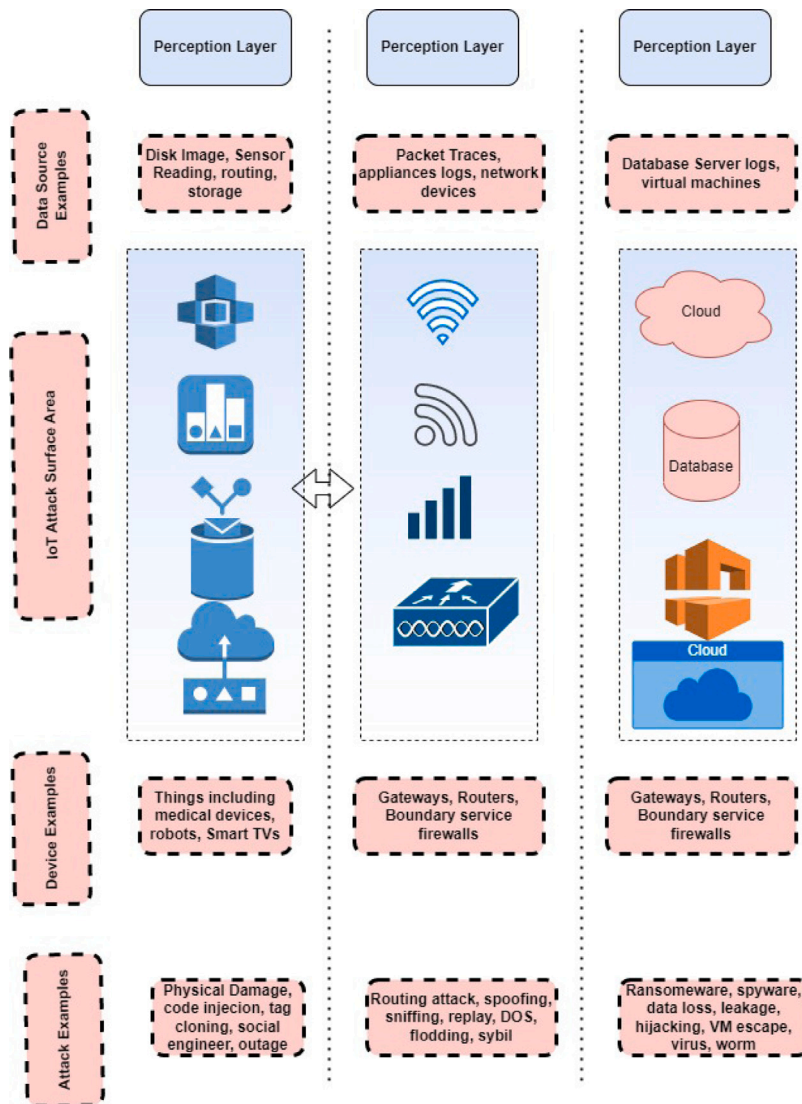


Fig. 8. IoT architecture and layer attacks.

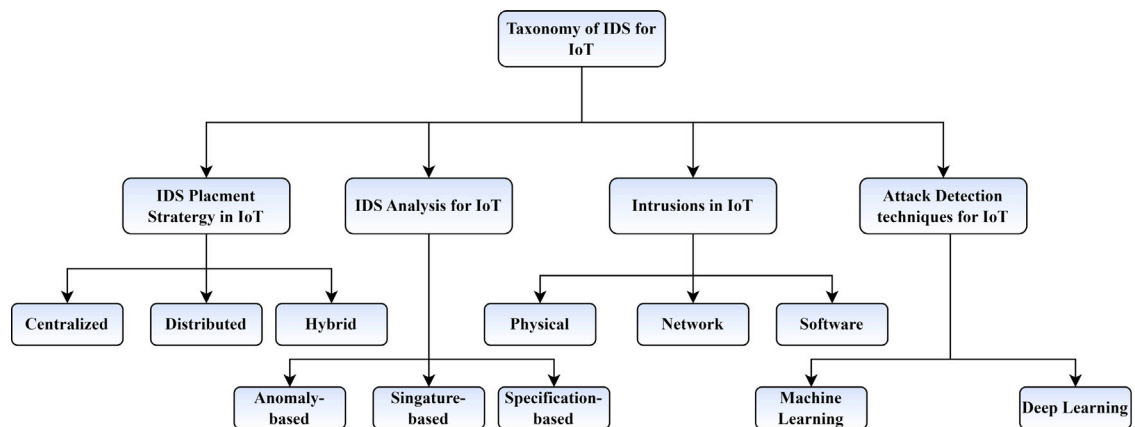


Fig. 9. Taxonomy of IDS in IoT technology.

Network intrusions. Network-based attacks exploit vulnerabilities in the communication layer, such as routing protocols or unsecured channels. Examples include:

- **Man-in-the-Middle (MITM) Attacks:** Attackers intercept and alter data during transmission.
- **Distributed Denial of Service (DDoS) Attacks:** Overloading IoT networks with traffic, as seen in the Mirai botnet attack.

Solutions involve:

- **Secure Routing Protocols:** Implementing protocols like RPL (Routing Protocol for Low-Power and Lossy Networks) with enhanced security features.
- **Traffic Filtering:** Using firewalls and anomaly-based IDS to detect unusual traffic patterns.

Software intrusions. Software vulnerabilities are exploited through malware such as viruses, worms, and ransomware. For instance:

- **Stuxnet Worm:** A well-known attack targeting industrial IoT systems.

Such malware can steal sensitive information, alter configurations, or delete critical files. Mitigation strategies include:

- **Regular Software Updates:** Ensuring devices run the latest firmware to patch known vulnerabilities.
- **Code Obfuscation and Sandboxing:** Techniques to prevent reverse engineering and isolate malicious software.

Encryption-based intrusions. Encryption attacks focus on breaking cryptographic algorithms or exploiting side-channel data to gain access to sensitive information. For example:

- **Side-Channel Attacks:** Observing power consumption or electromagnetic emissions to infer encryption keys.

Solutions include:

- **Advanced Encryption Standards (AES):** Using robust encryption algorithms with sufficient key lengths.
- **Noise Injection:** Adding noise to side-channel signals to obscure sensitive data.

IoT Security Challenges and Solutions

Continuous operation. IoT devices are designed to remain operational, making them susceptible to constant attacks. A proactive approach involves:

- **IDS:** Signature-based IDS like Snort and anomaly-based IDS using machine learning models (e.g., Random Forest, Support Vector Machines) to detect and mitigate threats in real-time.

Interconnectivity. The high degree of interconnectivity in IoT networks allows attackers to exploit one device to access the entire system. Solutions include:

- **Network Segmentation:** Dividing IoT networks into isolated segments to limit attack propagation.
- **Zero Trust Architecture:** Implementing strict access controls and continuous authentication.

Weak authentication and encryption. Many IoT devices lack robust authentication mechanisms and rely on weak encryption. Solutions include:

- **Public Key Infrastructure (PKI):** Using digital certificates for device authentication.
- **Multi-Factor Authentication (MFA):** Adding layers of security beyond passwords.

Ease of exploitation. IoT devices often have limited computational capabilities, making them easier to exploit. Solutions involve:

- **Lightweight Cryptographic Algorithms:** Algorithms like Elliptic Curve Cryptography (ECC) are designed for resource-constrained devices.
- **Secure Boot Mechanisms:** Ensuring only authenticated firmware is executed on devices.

IoT security is a critical concern due to the vast number of interconnected devices and their susceptibility to various attacks. Addressing these issues requires a multi-layered approach that combines robust encryption, network segmentation, secure communication protocols, and advanced intrusion detection systems. [Figs. 9 and 10](#) illustrate the taxonomy of IDS and the types of intrusions in IoT networks, providing a comprehensive overview of the attack vectors and their mitigation strategies.

4.1. Network-based intrusions

Network intrusions aim at attacking the IoT network by interrupting network activity. The following are examples of common forms of network attacks:

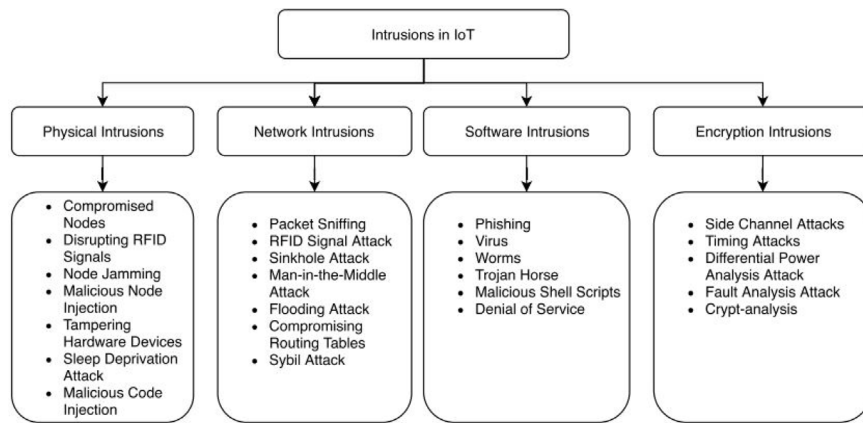


Fig. 10. Intrusions in IoT.

1. **Attacks Using Packet Sniffing:** Using packet sniffing, the intruder monitors network traffic as it flows. These programs sniff the data to get at the data in the packets' headers and payloads. The hacker will use this information to launch network breaches [218]. IoT networks use RFID, or radio-frequency identification technology, to keep tabs on the many devices connected to the network. This technology might be vulnerable to attacks such as those described below. RFID comprises two components: tags and signals for the connection between IoT devices. Spoofing, cloning, and other forms of unwanted access are all possible due to the vulnerabilities in the tags and signals. Spoofing is a sort of network intrusion in which a hacker fakes the transmission of legal signals to steal data from a target system [219]. In a cloning attack, an intruder generates a clone of a previously existing RFID tag to inject harmful information or obtain control of information flowing through the network [220]. An attacker can get access credentials by tampering with RFID signals and tags. This leads to illegal access to the information available at nodes of IoT networks. An intruder can eavesdrop, edit, or erase the information from the system.
2. **Sinkhole Attack:** To carry out this type of attack, the intruder first compromises a network node and then uses that node as a launchpad for other attacks. When this happens, the compromised node incorrectly reports the route distance to its neighbors. In this approach, it draws in traffic from the network and then either reroutes it or discards it. Computing the average hop count of each node in a network and comparing it to the average hop count is a certain way to spot a sinkhole attack [221]. For a particular node, it is possible to classify it as susceptible to sinkhole attack if the minimum hop count is less than a threshold number.
3. **Man-in-the-middle-attack:** Man-in-the-middle attacks occur when an adversary intercepts and sabotages data transmissions between two nodes. The attacker tries to intercept the network traffic between the victim nodes.
4. **Denial of Service (DoS):** With this type of attack, the attacker floods the nodes with requests for services. Consequently, targeted consumers are not able to acquire legitimate resources.
5. **Toxic Routing Tables:** In this scenario, the integrity of routing tables is broken by spoofing or otherwise modifying the routing information. This leads to a rise in the rate at which packets are dropped, the propagation of incorrect route information, and/or the partitioning of the network.
6. **Sybil Attack:** The attacker disrupts service on the network by assuming many fictitious identities belonging to the hacked node.

4.1.1.1. DS-RAM IDS

In the context of IoT security, **DS-RAM IDS** refers to an **Intrusion Detection System (IDS)** framework designed to enhance security in **Dynamic Source Routing (DSR)**-based networks while addressing vulnerabilities related to **Random Access Memory (RAM) management**. This type of IDS is particularly relevant in **resource-constrained IoT environments**, where efficient memory usage and robust intrusion detection are critical. DS-RAM IDS focuses on mitigating specific types of attacks that threaten IoT systems, ensuring secure communication and efficient operation even under constrained resources.

Sybil attacks. Sybil attacks are a significant threat to IoT networks, where a malicious node generates multiple fake identities to disrupt network operations, manipulate routing tables, or deceive IDS systems. These attacks can lead to false routing paths, interception of sensitive data, and even network partitioning. DS-RAM IDS mitigates Sybil attacks by employing lightweight cryptographic techniques to verify the identities of nodes. Additionally, it uses AI-based behavioral analysis, such as clustering algorithms, to analyze node behavior and identify inconsistencies that indicate Sybil attacks. This ensures that malicious nodes with fake identities are detected and isolated from the network.

Malicious node attacks. Malicious nodes within IoT networks pose a direct threat by disrupting communication, dropping packets, altering data, or injecting malicious packets into the network. Such attacks compromise data accuracy and disrupt the overall functionality of the network. DS-RAM IDS addresses this threat by incorporating trust-based models that evaluate the reliability of nodes based on their behavior. Furthermore, machine learning algorithms, such as Random Forest or Support Vector Machines (SVM), are utilized to classify nodes as malicious or benign. By identifying and isolating malicious nodes, DS-RAM IDS ensures the integrity and reliability of IoT networks.

Memory exhaustion (DoS) attacks. IoT devices, due to their limited resources, are particularly vulnerable to **Denial of Service (DoS)** attacks that aim to exhaust their memory. Attackers flood the network with excessive requests, causing the IDS or IoT devices to run out of RAM and fail. This can lead to system crashes, unresponsiveness, and an inability to detect further intrusions. DS-RAM IDS mitigates memory exhaustion attacks by implementing efficient memory management techniques to handle large volumes of data without overloading the system. It also employs AI-based traffic filtering to detect and block abnormal traffic patterns that indicate DoS attacks, ensuring the system remains responsive and functional.

Routing attacks (e.g., wormhole, blackhole). Routing attacks exploit vulnerabilities in the **Dynamic Source Routing (DSR)** protocol to disrupt data transmission and compromise network performance. In a wormhole attack, attackers create a tunnel between two malicious nodes to intercept and manipulate data, while in a blackhole attack, a malicious node drops all packets passing through it. These attacks can severely degrade the performance and reliability of IoT networks. DS-RAM IDS addresses these threats by enhancing the DSR protocol with secure route validation mechanisms. Additionally, it uses AI-driven anomaly detection to identify deviations in routing behavior, such as unusually short paths indicative of wormhole attacks or high packet drop rates characteristic of blackhole attacks.

Side-channel attacks. Side-channel attacks are sophisticated threats where attackers exploit side-channel information, such as power consumption or timing data, to infer sensitive information like encryption keys. These attacks compromise the confidentiality of IoT communications and can lead to severe security breaches. DS-RAM IDS mitigates side-channel attacks by incorporating noise into side-channel signals, making it difficult for attackers to extract meaningful information. It also employs advanced encryption standards (AES) to ensure robust encryption, protecting sensitive data from cryptographic attacks.

AI techniques for intrusion detection. To effectively detect and mitigate these diverse threats, DS-RAM IDS leverages advanced AI techniques. Clustering algorithms are used to detect Sybil attacks by identifying anomalous patterns in node identities. Supervised learning models, such as Random Forest, SVM, and Neural Networks, classify malicious nodes and detect routing anomalies. Reinforcement learning is employed for adaptive intrusion detection, allowing the IDS to learn and improve its detection capabilities over time. Additionally, deep learning techniques are utilized to analyze complex attack patterns, such as side-channel data or large-scale DoS attacks. These AI techniques ensure that DS-RAM IDS remains effective in detecting and mitigating a wide range of intrusions in resource-constrained IoT environments. By narrowing its scope to specific types of intrusions, such as Sybil attacks, malicious nodes, routing attacks, and memory exhaustion, DS-RAM IDS provides a robust framework for securing IoT networks. Its integration of AI techniques ensures efficient detection and mitigation of these threats, even in resource-constrained environments. Future research should delve deeper into optimizing DS-RAM IDS for emerging IoT challenges, such as edge computing and 5G-enabled networks, to ensure comprehensive security.

4.2. Physical intrusions

The hardware components of an IoT network are the target of physical invasions. These are some examples of common forms of physical attacks against IoT networks.

1. **Insecure Nodes:** The attacker manipulates a node by either replacing it with a malicious one or altering its configuration. Information can be stolen and access denied if an intruder can modify the routing tables or other sensitive data. Data processing in the higher levels of an IoT network is impacted by communications that pass via a hacked node [222].
2. **Interfering With Radio Frequency:** Identification Signals When the RFID reader has been compromised, the RFID systems are hampered. Transmission of spurious signals, rather than desired radio signals, compromises the reader component. Because of this, wireless IoT network devices experience communication disruption [223].
3. **Contamination of Nodes and Jamming:** One type of exploit used to cause a denial of service in wireless networks is node jamming, in which legitimate network communication is blocked from reaching authorized devices by the use of jamming software that sends out frequencies of unauthorized traffic. Changes to the transmission frequencies in the network have rendered it inoperable [224].
4. **Infusion of a Malicious Substance into a Node:** An attacker, physically inserts a malicious node in the network. This node snoops on legitimate network traffic and steals information from its victims. All data flowing over the network might be vulnerable to manipulation by the rogue node. An attacker injects a malicious node into the network by producing a duplicate of the node, rendering the victim node incapable of sending or receiving packets [225].
5. **Hardware Component Tampering:** To launch a denial-of-service attack (DoS), an attacker might cause damage to the devices and components of an IoT network [61].
6. **Manipulation of Society** One method of attacking the IoT is by direct physical contact with the devices themselves. As a result, the attacker has access to private system data [226].
7. **Sleep Deprivation Attack:** An attacker uses this technique to reduce the lifespan of nodes in a network by increasing their power usage [227].
8. **Injecting Malicious Code:** To compromise an IoT system, an attacker injects malicious malware into it.

4.3. Software intrusions

The software incursions tamper with the data contained in the system resources and even affect processing operations in the system. The system intrusions comprise hostile software programs in the form of viruses, worms, malware, or Trojan horses. Different forms of software invasions are as follows

1. *Phishing*: The spoofing of email addresses is used in this attack. Malicious programs are frequently sent as attachments through email. The goal of a phishing attack is to get sensitive information, such as a user's password or bank information [228].
2. *Infectious Programs*: Many different types of malware can do this. An attacker can use this harmful code to cause damage to the system's resources. Code is introduced into the system through downloadable files and email attachments. When a user downloads the file or opens the attachment, the virus is triggered. A worm is a malicious application that may spread to other computers without any intervention from the user. Anti-virus, firewall, and IDS are used to identify these harmful applications. In Alani et al. [229], for instance, a hybrid IDS is created using anomaly-based and signature-based IDS with a honeypot to identify worms in the system.
3. *Shell Scripts with Harmful Code*: To acquire access to local user system resources, malicious shell scripts are injected into a network remotely. This gives the attacker complete remote control over all user data and processes. In addition, the attack may read and change any files on the system [230].
4. *Service Defeat* This exploit prevents authorized users from accessing the system's features. The target of this attack is the application itself. Users continue to be denied access to the services because the apps are unable to handle the volume of requests [231].

4.4. Encrypted intrusion

Encryption attacks center around interfering with or stealing the public and private encryption keys, which compromises the encryption process. Below are only a few examples of attacks that have been made against various forms of cryptography.

1. *Channeled Side Attack*: Here, an attack is carried out or encryption keys are stolen by using side channel data from encrypted devices. Information on power use, processing time, and error rates may be found in the side channel data. To steal encryption keys, this data is utilized. Timing attacks, differential power attacks, and fault analysis attacks are all examples of side-channel attacks [232]. The time it takes a computer to accomplish a task is exploited in timing attacks. Stealing the secret keys, destroying cryptosystems, or stealing information on the components of encryption methods like Diffie–Hellman exponents or RSA keys is all possible with this knowledge.
2. *Invasion through Cryptanalysis*: An adversary uses either plaintext or encrypted message to steal the cryptographic keys in a key exchange attack. Several types of cryptanalysis attacks, such as the ciphertext alone attack, the known plaintext attack, the selected ciphertext attack, and the chosen plaintext attack [233], are possible depending on the method employed. An attacker conducting a ciphertext-alone attack relies solely on the ciphertext itself to deduce the plaintext. For a known plaintext attack, an intruder must know part of the plaintext to decrypt the encrypted text. In the selected plaintext and chosen ciphertext attack, the attacker opts for plaintext.

5. Machine learning-based IoT security

Many domains of IoT applications and systems make use of Machine Learning and Deep Learning methods. Fig. 11 shows the overview of different services in the current domain. To further understand how ML and DL may be used in an IoT system, consider the diagram depicted in Fig. 4. Due to their ability to improve with practice, these methods are increasingly being deployed for use in IoT network attack classification. ML and DL methods have become increasingly significant as a result of the proliferation of new algorithms, the production of massive amounts of data, and the decline in processing costs [234]. ML and DL methods have progressed over time in conducting an empirical analysis of a wide range of applications. This research explores the use of ML and DL approaches to the problem of identifying and categorizing attacks in IoT networks. With the use of ML methods, characteristics that are useful for classifying attacks on IoT networks may be extracted through a process called feature engineering. In contrast, DL methods accomplish pattern analysis by abstracting discriminative or generative characteristics through several layers of linear and non-linear processing [235]. The goal of this discussion of ML and DL methods for safeguarding IoT networks is to offer a comprehensive understanding of these methods. Algorithms for ML and DL use a dataset to solve a problem in a specific application. The data set is divided into a training set and a test set. Many dataset features may be learned and examined with the use of the study set, which is the training set. In the case of intrusion detection, for instance, algorithms may be taught to use attributes from a training dataset to determine if a particular data sample represents an attack or is typical. Machine learning and deep learning algorithms' job is to enhance the system's classification precision through the study of network traffic patterns under both normal and attack conditions.

There are two main types of ML algorithms: classification and clustering. By assessing input factors and connecting them with predicted output, classification algorithms construct prediction models from labeled data samples. Hence, the parameters of both the input and the output are linked using these techniques. Training a learning model using a given training set is performed during the classification algorithm's training phase. The training phase is followed by a prediction and classification phase [236], using

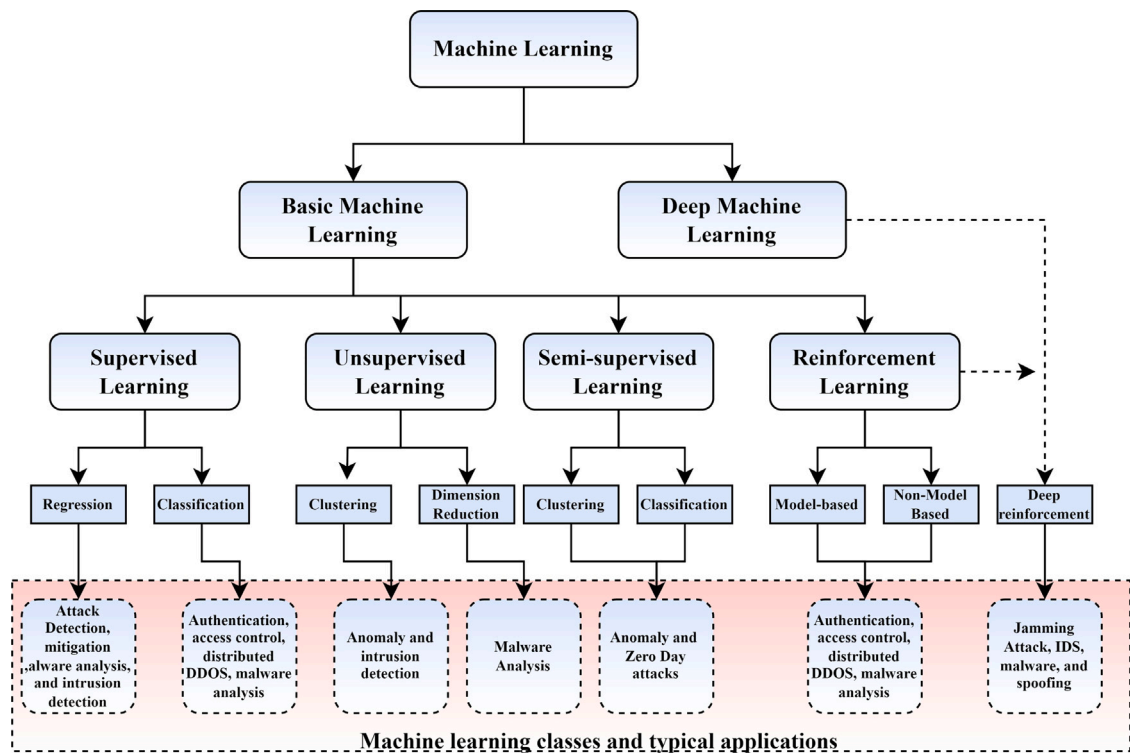


Fig. 11. Machine learning-based security services.

the knowledge gained. In supervised DL methods, a multi-layer network is constructed with an input layer, an output layer, and optionally one or more hidden layers. Neurons serve as nodes in these tiers, with edges connecting the various tiers. Neurons are given arbitrary weights to start, and the input parameters are multiplied by those weights to produce an output [237]. Clustering methods are well-known for learning from unlabeled datasets, whereas classification methods are well-known for learning from represented and labeled data. While traditional machine learning approaches need labeling data before they can be used for training, clustering methods may learn from unlabeled data by grouping it into discrete clusters based on their similarities. Reinforcement Learning (RL) is another typical ML method [238]. Cooperative learning (RL) methods use the surrounding environment to acquire new information. The purpose of this strategy is to examine environmental factors and choose the most effective approach for dealing with environmental agents [68]. RL methods are essentially trial-and-error procedures. A set of behaviors is defined for a given set of input parameters based on environmental characteristics. The benefits, drawbacks, and uses of ML and DL methods for protecting IoT networks are covered here.

5.1. Machine learning technique for IoT network

The benefits, drawbacks, and uses of the most popular ML algorithms, including decision trees (DT), support vector machines (SVM), Bayesian algorithms, k-nearest neighbor (KNN), random forest (RF), association rule (AR) algorithms, ensemble learning, k-means clustering, and principal component analysis (PCA), are discussed.

5.1.1. Decision trees

Samples of data are sorted into categories in the Decision Tree (DT) classifier according to the values of their features. Here, information is laid out in a tree structure, with each leaf representing a different decision rule that divides the data based on feature value. The goal of the DT classifier is to learn decision rules from the training data that may be used to assign classes to the target variable. Information gain and the index are used to determine the best collection of features to employ [239]. Building the decision tree and classifying data are the two fundamental tasks of a DT classifier. The decision tree's starting point is determined by summing the information gains of all the features in the dataset, also known as the Gini index. The top-scoring label class is chosen as the root node based on the Gini index or gain value. The decision tree is constructed by progressively computing gain values and the gini index for additional class labels. This continues until the decision tree's feature nodes are labeled with their corresponding classes. Following tree construction, a feature set for the new samples may be determined by following the tree's nodes until reaching a leaf node. Here is how fresh input samples may be anticipated by the class [240]. There are many applications for decision tree models, including [variable selection], [variable relative importance], [missing value management], [predicting class labels], and [data manipulation].

1. **Node Type:** Node Construction Root nodes, intermediate nodes, and leaf nodes are the three types of nodes that make up a decision tree. Both the outside nodes and the inner nodes of a specific dataset's tree indicate the classes that make up that dataset. The gain index value of a set of class labels is used to create these.
2. **Edge:** The connection between the root node and child nodes represents the desired decision or outcome. Each branch leading from the ancestor node or an intermediate node to a leaf node represents a decision rule for how the data should be sorted.
3. **Node Partitioning:** To separate The internal nodes are partitioned into leaf nodes based on input variables of the identified target variables. Both discrete and continuous inputs are accepted. Unless the predetermined stopping requirements are met, node splitting will continue.
4. **Early Assessment:** The analysis and characteristics of the dataset are taken into account while deciding on a termination condition. Termination conditions are often determined by factors like the minimum amount of data samples, the minimum number of records in a node before splitting, and the number of levels away from the root node [241]. To achieve robustness and simplicity, the DT model defines the termination criteria.
5. **Pruning:** Pruning is done to reduce the size of a huge DT by getting rid of unnecessary nodes or just revealing as much as is necessary [242].

Securing IoT networks and devices against breaches involves the use of decision trees (DT) in conjunction with various machine learning (ML) classifiers [243]. For example, [244] proposes a fog-computing-based intrusion detection system (IDS) to protect VPN-connected IoT devices from distributed denial-of-service (DDoS) attacks. In this approach, DT is utilized to detect DDoS attacks by identifying the origins of malicious traffic. For network anomaly detection in IoT devices, [245] employs DT to construct behavioral profiles of these devices. [246] introduces a hybrid approach combining inverse weight clustering and DT to develop an IDS for IoT-based systems. The inverse weight clustering algorithm groups data based on similarities, and DT assigns classes to each group. Challenges such as inefficient use of computational resources arise when unnecessary data is processed or flagged for disregard. Reducing the false positive rate is therefore critical for improving the effectiveness of IDS. Combining support vector machines (SVM), DT, and Naive Bayes (NB), as demonstrated in Chando et al. [247], helps reduce false positive rates, creating a more effective IDS. In Chauhan et al. [248], a hybrid strategy for IoT network intrusion detection is described. This framework combines a misuse-based detection approach with an anomaly-based detection mechanism. DT is applied in the misuse-based model for attack detection and classification, while anomaly-based intrusion detection is performed using SVM.

5.1.2. Support vector machines

Support vector machines (SVM) classify data by constructing a hyperplane to divide it into two or more groups. Data points are arranged to maximize separation from the hyperplane, which passes through them. The generalization properties of SVM make this approach suitable for datasets with high feature counts but low sample sizes [249]. Originating from statistical learning [250], SVM was initially used for the two-dimensional classification of linearly separable data. With the use of kernel tricks [251], SVM can also classify and separate non-linear data by transforming it into a linear form before applying the hyperplane. SVM has been applied to train on data and understand attack patterns for real-time intrusion detection in IoT networks. For instance, ElDahshan et al. [252] developed a kernel-based SVM technique for network-based intrusion detection by combining the radial basis function kernel, the polynomial kernel, and game theory. Attique et al. [253] employed SVM to construct IDS for mobile ad hoc networks, identifying and eliminating intruding nodes. The proposed method demonstrated independence from routing protocols, node mobility, and network size.

Multi-class classification in IoT networks using SVM with chi-square feature selection is explored in Alsulami et al. [254]. The temporal complexity of SVM, measured as $O(N^2)$, where N is the number of data points, ensures both space efficiency and speed, making it suitable for various security applications [255]. For protecting IoT network infrastructure from malware, [256] proposes a linear SVM-based Android malware detection framework. This framework was compared with other ML methods such as NB, radial basis function (RF), and DT, with findings favoring SVM. Similarly, Liu et al. [251] introduced a framework for detecting Android malware in IoT devices using automatic learning techniques. However, the effectiveness of SVM varies depending on datasets, environments, and attack scenarios. SVM has also been used to protect smart grid infrastructure, as demonstrated in Anzi et al. [257], which highlighted its efficacy in identifying both common and uncommon forms of attack. Additionally, experimental studies, such as those in Hussain et al. [258], show that SVM can be exploited to breach IoT device security, enabling attacks like template attacks.

5.1.3. Naive Bayes

The Naive Bayes (NB) method calculates the posterior probability of an event using Bayes' theorem [259]. This approach is applied in intrusion detection to analyze specific aspects of network traffic and identify probing attacks. The prior probability of a defined class for a dataset is used to investigate the occurrence probabilities of various attacks and normal traffic. Classification using NB involves calculating the likelihood that a given input sample's features belong to a specific class. For IDS, NB uses network traffic characteristics, such as connection duration, protocols, and connection status flags, to classify data samples as benign or abnormal. Although these characteristics may be interdependent, NB treats them independently, as each feature equally influences the sample's likelihood.

Aldarwbi et al. [260] employed NB alongside other ML approaches to minimize false positives in intrusion detection. NB is favored for its simplicity and ability to handle binary and multi-class classification, even with irrelevant features and limited training data [261]. However, NB cannot establish meaningful connections between features and input variables, which may limit

its performance. Ramana et al. [262] proposed a methodology for preventing DDoS attacks on IoT infrastructure, where IDS agents use NB to detect suspicious traffic and actions at individual nodes. Alars et al. [263] introduced a two-tier classification module combining NB and k-NN with certainty factors to identify U2R and R2L threats. Abdoli et al. [264] used NB and SVM to identify image spam by transforming images into text and analyzing the extracted text area for spam classification.

5.1.4. K-nearest neighbor

The k-nearest neighbor (k-NN) method is a non-parametric approach for classification and regression problems. It classifies data based on the k closest training samples in the feature space, with the parameter k being user-defined. Training data is treated as vectors in a multi-dimensional feature space, each associated with a class label [265]. During classification, class labels are ranked using distance measures such as Euclidean distance for continuous variables and Hamming distance for discrete variables. Weighted classification techniques can address skewed class distributions by adjusting the importance of each k-NN data point based on its distance.

Feature selection or extraction algorithms are often employed to improve k-NN performance. Yaseen et al. [266] used k-NN for intrusion detection in IoT networks, focusing on dimensionality reduction and classification of U2R and R2L attacks. Khonde et al. [267] proposed a framework to detect attacks by analyzing energy consumption statistics of monitored devices. A two-phased k-NN and neural network approach reduced computational time while maintaining accuracy. Balasaraswathi et al. [268] developed an IDS for mobile ad hoc networks to identify malfunctioning nodes and improve packet delivery ratios.

5.1.5. Random Forest

Random Forest (RF) is a classification method that constructs multiple decision trees and aggregates their predictions to determine the final output. RF avoids overfitting by using subsets of rules for classification, making it more robust than standard DT methods. RF also uses mutual information to prioritize features for classification. However, generating large decision trees can be computationally expensive, limiting RF's efficiency with large datasets.

Wu et al. [269] applied RF for DDoS attack detection in IoT networks, demonstrating superior performance compared to other ML methods such as SVM, k-NN, and ANN. Apruzzese et al. [270] employed RF to identify unauthorized IoT devices by analyzing network traffic characteristics. Gu et al. [271] combined RF with SVM for feature selection and attack classification in IDS.

5.1.6. Association rule mining

Association rule (AR) mining identifies relationships between variables in datasets to construct prediction models. Gao et al. [272] used AR to analyze TCP/IP parameters and detect various attack types. Gowdhaman et al. [273] applied AR techniques for intrusion detection, developing models based on fuzzy association rules. However, AR methods are less commonly utilized in IoT security due to their high computational complexity, which increases with the number of variables. Combining AR with other methods can enhance its performance and applicability.

5.1.7. Ensemble learning

The goal of ensemble learning in ML is to enhance classification accuracy and detection rate by pooling the results of many classification techniques. To construct the prediction model, EL utilizes a mixture of homogeneous and heterogeneous classifiers. The classification of data streams is an area where EL techniques shine. This is because dynamic modifications, such as the removal of a functionality or the addition of a different classifier, may be quickly included in the framework. When it comes to practicality, ML methods vary greatly depending on both the nature of the dataset being utilized and the specific task at hand. Hence, the approach utilized for one application dataset may not be suitable for another. Even more so, EL guarantees superior predictive performance over the single classifier model. To this end, EL is employed by merging various classifiers to improve the system's precision. As it draws from several different approaches, EL can reduce variability and accommodate over-fitting data with ease. Better results may be produced by the EL-based system using the provided set of hypotheses, and the system also shows signs of flexibility. These approaches, which utilize a network of classifiers, are superior to those employing a single classifier, although they are computationally intensive.

Intruder and virus detection have both made use of EL. Examples include [274], where an ensemble of tree classifiers is used to identify network attacks in IoT networks. In this case, the suggested architecture employs REPTree as the primary classifier alongside a bagging ensemble to identify and categorize attacks. It is recommended to use a combination of SVM, k-NN, and Particle Swarm Optimization (PSO). In this case, the proposed framework employs PSO to allocate weights to an ensemble of support vector machines (SVMs) and k-nearest neighbors (NNs) for attack classification. Similarly, SVM is used with rough set theory to better extract characteristics from the dataset that may be used in the detection process. In Koyama et al. [275], ensemble learning is employed for malware detection on Android devices. This dataset was collected from McAfee's internal repository, and the malware was detected using an ensemble of additive logistic regression and RF classifier. In addition, a lightweight, application-agnostic, ensemble architecture is proposed for detecting abnormalities in an IoT network, which significantly reduces the computing time required in a wireless network with IoT devices operating under severe resource constraints. In this study, two problems are tackled at once: (i) develop an automated technique for intrusion detection in networks with limited resources, and (ii) assess the effectiveness of this method with a real-time dataset. The outcomes demonstrated that the ensemble-based architecture outperformed the separate classifiers.

Table 3

Comparative analysis of machine learning techniques for IoT.

Reference	Contribution	Attack type	Technique	Dataset	Results
[278]	Fuzzy association rulesets are exploited for attack classification	DoS, R2L U2R	Association Rule	KDD CUP	Accuracy-91.2%
[279]	Clustering for refining the output	DoS, R2L U2R	SOM	KDD CUP	Detection Rate- 89.69%
[280]	Detection of android malware	Malware	SVM	Simulated dataset	Accuracy-99.2%
[281]	Using ensemble learning for feature extraction	Malware	Random Forest	McAfee's internal repository	Accuracy-98.5%
[282]	Implementing incremental learning	Anomaly	Fuzzy System	Intel lab	Precision-95.6%
[283]	Two-tier classification is performed by combining NB	DoS, Probe, U2R, R2L	NB, k-NN	NSL-KDD	DR: 84.82%
[284]	A lightweight boolean masking countermeasure	Side channel attack	SVM	DPAContest V4	Accuracy-89.2%
[285]	Using genetic fuzzy systems for attack classification	DoS, R2L U2R	Association Rule	KDD CUP	Accuracy-99.2%
[286]	Integrated cluster with highest connectivity packet dropping algorithm	Anomaly	K-NN	Simulated dataset	Performance-85.6%

5.1.8. K-means clustering

Clustering techniques like the k-means algorithm divide information into subsets that share commonalities. The value k indicates how many distinct groups this method will generate. For each iteration, each data sample in the dataset is placed into one of the clusters depending on their similarity index. For this reason, the k-means algorithm employs an iterative process to get the best possible classification for a fresh input sample. Here, in the k-means method, the number of clusters and the data samples from the dataset are input to the training phase. The k-means method works by first assuming the k centroids, and then using the centroids to calculate the Euclidean distance between each data sample and the original. The data points are put into the cluster that is the closest to the centrifugal center. When the data samples are clustered, the cluster centers are recalculated by averaging the samples in that cluster. The procedure repeats itself until the termination criterion is met, i.e., until no data sample can update the current cluster center. The selection of k before the start of the k-means clustering process is its main problem. This approach also implies that each cluster has the same amount of data points. Computing a feature similarity index is another use for the k-means method in intrusion detection. K-means and DT [276] were used to construct IDS architecture for monitoring network behavior and issuing warnings for any hostile activity identified. A clustering algorithm's effectiveness with an unlabeled dataset is well-established. IoT system security has seen the application of k-means clustering to WSNs. A multi-kernel technique for Sybil attack detection in industrial WSNs is presented. To identify genuine sensors from Sybil ones, channel vectors are clusters in this case. A clustering-based method is described in Data et al. [277] for protecting user privacy in IoT networks. Clustering algorithms are used to boost the network's data exchange security. Table 3 provides a comparative analysis of ML methods applicable to the IoT. It can be seen that numerous research works have been proposed for determining attack categories using ML techniques. Moreover, the significant aspect is the performance registered by each of the works for the detection of the attack.

5.1.9. Deep learning methods for IoT security

To fully realize the potential of the IoT (IoT), it is necessary to use DL approaches. Because of their superior performance on huge datasets, DL approaches have largely supplanted ML ones. Because of the sheer volume of data produced by the IoT devices in use in the network, DL methods are well suited for this setting. In addition, DL methods can improve data representation by extracting characteristics from it. Moreover, IoT network devices may talk to one another and with applications without any intervention thanks to the layered design of DL approaches. For instance, automating device interactions via the IoT is essential for constructing a smart home. The computational architecture of DL methods is multi-layered, with each layer serving to develop increasingly abstract representations of the underlying data. DL methods, as opposed to classic ML methods, may be applied to much-improved software. DL methods, which are a subset of ML methods, employ many layers of non-linear processing to get their results. These processing layers extract and represent features, either generative or discriminative, for learning patterns in data. DL methods have a hierarchical representation of data thanks to their deep architecture. Hence, DL techniques are sometimes called hierarchical learning. The structure of DL methods is based on that of the human brain, which uses networks of interconnected neurons to process information. Two types of DL methods are distinguished: those that seek to discriminate and those that want to generate. Discriminative DL methods include Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs),

Table 4

Comparative performance analysis with related deep learning-based security works.

Ref	Methods	Accuracy	Recall	Precision	FAR	FPR	F1-Measure	FNR
[156]	AE	98.95%	N	92.15%	N	N	N	13.25%
[287]	LSTM	97.59%	N	N	N	N	94.15%	N
[288]	LSTM + RNN	94.56%	N	93.14%	N	N	N	14.15%
[289]	AE	96.25%	96.75%	N	93.25%	94.15%	96.25%	N
[290]	DBN	96.14%	N	94.15%	N	96.17%	N	N
[291]	VAE + DNN	94.59%	94.15%	93.33%	N	94.14%	N	18.25%
[292]	CCN + VAE + LSTM	93.26%	N	92.15%	N	93.21%	N	N
[293]	DBN	99.25%	N	94.15%	N	94.15%	94.15%	N
[294]	CNN	92.23%	94.16%	N	N	N	N	14.55%
[295]	CNN	93.25%	N	N	N	N	N	15.14%
[296]	AE	91.26%	93.15%	N	N	96.21%	93.21%	N
[297]	CNN + LSTM	93.25%	N	N	N	N	N	N
[298]	CNN	95.15%	N	N	N	93.33%	90.25%	15.15%
[299]	Multi-Layer	94.15%	N	N	N	N	N	N
[300]	deep learning	93.25%	N	N	N	94.15%	89.25%	N
[301]	DFFNN	92.25%	N	N	N	N	N	N
[302]	LSTM	93.26%	N	N	N	N	N	N
[303]	CNN	93.26%	N	89.25%	N	N	N	N
[304]	CNN	94.15%	N	N	N	N	N	2.2%
[305]	CNN + AE	94.16%	N	N	N	89.25%	N	N
[306]	AE	96.17%	N	N	N	N	N	N
[307]	AE	96.15%	N	89.25%	N	N	N	3.25%
[62]	GNN/RNN	95.15%	N	N	N	N	N	N
[308]	CNN + AE	94.45%	N	N	N	89.25%	N	3.15%
[309]	AE	94.44%	N	N	89.25%	N	N	N
[310]	DML	96.25%	N	N	N	N	N	N

whereas generative DL methods include Deep Autoencoders (AEs), Deep Belief Networks (DBNs), Restricted Boltzmann Machines (RBMs), and Generative Adversarial Networks (GANs). The employment of multiple DL methods in concert is also common, and this method, known as Ensemble DL, is the term for it. Table 4, Table 5, and Table 6 present the comparative analysis with a state-of-the-art literature review for performance assessment.

5.1.10. Convolutional Neural Networks (CNNs)

To restrict the data properties employed by a conventional neural network, Convolutional Neural Networks (CNNs) were developed. Sparse interaction, parameter sharing, and equivariant representation were used to reduce the amount of data characteristics needed. The increased computational processing and lower training time complexity arise from fewer connections between layers. CNNs are built with a convolutional layer and a pooling layer. Several kernel functions of the same size are used to process data properties in the convolutional layers. There are two distinct kinds of pooling layers, called maximum and average, respectively. To make the next set of layers more manageable in size, sampling is done in this set of layers. The max-pooling layer takes the input data and selects the largest value from each non-overlapping cluster. On the other hand, in an average pooling layer, values are averaged over clusters. Each character in the feature space is subjected to a non-linear activation function in CNN's activation layer. Learnability and the availability of non-linear functional mappings between input parameters and response variables are greatly aided by activation functions. Sigmoid, tanh, and the Rectified Linear Unit Activation (RELU) function are only a few examples of activation functions. CNN's automated feature learning from the dataset is what ultimately leads to its superior accuracy, so it is easy to see why it is so widely used. However, the computational cost of CNN algorithms is rather expensive, making it difficult to implement them in networks with limited resources [4,5].

A distributed architecture can be used to solve this problem. For a decentralized system, it constructs a lightweight deep neural network with just a small number of output classes. To achieve deep classification, network training is done in the cloud. The use of CNN methods in image identification applications has gained widespread acceptance. CNN makes use of the massive publicly available resource Imagenet to demonstrate how well CNN can be utilized for picture classification and identification. CNN is used to categorize distant sensing data to protect wireless sensor networks. In addition, CNN is utilized to safeguard IoT devices from android malware. Here, a Convolutional Neural Network (CNN) technique analyses malware-related raw data and automatically learns characteristics for detecting malware. So, with the CNN method, the feature engineering procedure is not required. So, during training, CNN automatically learns the features and classifies the incoming data at the same time. The necessity for feature selection and feature extraction strategies to construct an efficient classification model [141] is therefore removed. Furthermore, intruders exploit this CNN learning process as a tool to bypass protections. In Gowdhaman et al. [273], for instance, CNN is used to subvert the network security measures that have been put in place, including cryptographic functions and protocols.

5.1.11. Recurrent Neural Network

A Recurrent Neural Network (RNN) is a type of DL approach that uses directed graphs to model the relationships between data points. Using temporally consecutive data, it displays dynamic behavior. Whereas other types of neural networks rely on external

Table 5
Comparative analysis for accuracy range of deep learning.

Ref	Method	Accuracy range
[311]	CNN	75.56–92.26%
[312]	CNN + AE	99%–100%
[133]	AE	80–92.26%
[313]	CNN	98%
[314]	CNN + LSTM	99.15%
[315]	CNN + AE	85.69–95.56%
[316]	LSTM	95.65%
[317]	LSTM	95.56%
[28]	CNN + LSTM	95.15%
[318]	DBN	94.15%
[319]	LSTM + RNN	98.02%
[320]	GNN/RNN	98.56%
[321]	CCN + VAE + LSTM	70.15–91.25%
[322]	VAE + DNN	82.16%
[323]	DFFNN	99.25%

Table 6
Comparative analysis of tools and simulators for IDS security.

Ref	Methods	Keras	Tensor flow	PyTorch	Scikit	SoftMax	R-tool	Cooja	Raspberry	Hybrid	Python	Sigmoid	MATLAB
[324]	AE	Y	N	N	N	N	N	N	N	N	N	N	N
[325]	LSTM	N	Y	N	N	Y	N	Y	Y	Y	N	N	Y
[326]	LSTM	Y	N	N	Y	N	N	N	N	Y	Y	Y	N
[327]	VAE + DNN	N	N	Y	N	Y	Y	N	Y	N	N	N	N
[133]	LSTM	Y	N	N	Y	N	N	N	N	Y	N	N	Y
[328]	CNN + LSTM	N	Y	N	N	Y	N	Y	Y	N	N	N	Y
[329]	VAE + DNN	Y	N	Y	N	N	N	N	N	N	N	N	N
[330]	AE	N	N	N	N	N	N	N	Y	N	N	Y	Y
[331]	DBN	N	N	N	Y	N	Y	N	N	N	Y	N	N
[332]	VAE + DNN	Y	N	N	N	N	N	N	N	N	N	N	N
[333]	CCN + VAE + LSTM	N	N	N	N	N	N	Y	N	N	N	N	Y
[334]	CCN	N	N	N	Y	Y	N	N	N	N	N	N	N
[335]	DBN	N	N	Y	N	N	N	N	N	N	Y	Y	N
[336]	AE	Y	Y	N	Y	Y	N	Y	N	N	N	N	N
[337]	DBN	Y	N	N	N	N	N	N	N	Y Y	N	N	N
[338]	CNN + LSTM	N	Y	N	Y	N	N	Y	N	N	N	N	N
[339]	CCN + VAE + LSTM	Y	N	Y	N	Y	N	N	N	Y	N	N	N
[340]	CNN + LSTM	N	N	N	N	N	N	Y	N	N	N	Y	Y
[341]	LSTM + RNN	N	Y	N	N	N	N	N	N	N	N	N	N
[342]	CNN + LSTM	Y	N	N	Y	N	N	N	N	N	Y	N	Y
[343]	LSTM + RNN	N	N	N	N	N	N	N	Y	N	N	N	N
[344]	DBN	N	N	N	N	Y	N	N	N	N	Y	Y	N
[345]	CNN + AE	Y	Y	N	N	N	N	Y	N	N	N	N	Y
[346]	DBN	N	N	N	N	N	N	N	N	N	N	N	N
[347]	CNN + AE	N	N	N	N	N	N	N	N	N	N	N	Y
[348]	LSTM + RNN	Y	N	N	N	N	N	N	N	Y	N	N	N
[349]	CNN + AE	N	Y	N	N	N	Y	N	N	N	N	Y	N
[350]	GNN + RNN	N	N	N	N	N	N	N	N	Y	Y	N	Y
[351]	DBN	N	Y	N	N	Y	N	N	N	N	N	N	N

memory to process inputs and remember their prior state, RNNs have their internal memory. Application domains where RNN methods have been used include handwriting recognition, text classification, and wireless sensor networks. In RNN, “recurrent” refers to networks with a structure that includes both finite and infinite impulses [352]. Both social and professional networks evolve. Networks with finite impulse forms are directed acyclic graphs, whereas networks with infinite impulse forms are directed cyclic graphs, but other than that, they are identical. More storage capacity for tracking the evolution of data sequences is available in both varieties of networks. Long-Short Term Memory (LSTM) and Gated Recurrent Units (GRU) are examples of RNNs that combine a feedback loop and a storage space with a time delay approach, respectively. RNN can be helpful in situations when the result is derived from examining historical data. Now, the network’s output is conditional on the values that were fed into it before. Because there is no correlation between the input and output layers, a feed-forward neural network cannot be used with this kind of information [353]. One of the key components of RNN design is the temporal layer, which is used to detect and learn from time-dependent differences in complex sequences of data. The units of the recurrent nodes conceal this information. Using the data already in the network, the concealed nodes are updated constantly to reflect the current state of the network. The ability of RNNs to efficiently manage sequential data has led to its widespread use. This aspect of RNN is helpful in threat detection since threat patterns change over time. Hence, RNN may be used to boost IDS efficiency.

Unfortunately, RNN networks have problems with disappearing and ballooning gradients. With RNNs, IoT infrastructure can be protected, and all its connected devices. Data in vast amounts are produced sequentially by IoT network devices. Key characteristics for pinpointing any flaws in the network are gleaned from information gathered through network traffic flows and interactions between devices. In Guan et al. [354], for instance, RNN is used to identify botnet attacks via monitoring network activity. In Jose et al. [355], RNN is used to identify malfunctioning nodes in WSNs. A dense recurrent neural network (RNN) is utilized to spot intrusions in a domestic IoT setting in Alshammari et al. [356]. In Mishra et al. [357], RNN is used to identify malware on IoT devices. Protecting IoT (IoT) systems and devices from time-series attacks [358] is a use case for RNN and its variations.

5.1.12. AutoEncoders

The autoencoder is an unsupervised DL method for learning new data [86]. The primary goal of AE is to encode the provided data while simultaneously decreasing the dataset size by reducing noisy signals. Data dimension reduction is accompanied by an attempt to recreate a representation that is faithful to the original input. Spatial, denoising, recursive, stacking, and contractive autoencoders are only a few examples of the many varieties utilized for efficient classification. Encoder and decoder functions make up the entirety of an AE network. Encoding is the process of taking a data sample as input and producing a new data abstraction, or code, as output. Decoding uses the encoded data as input to attempt to reconstruct the original message. Reconstruction errors are therefore minimized and the training phase of AE is concluded.

AE is restricted by the need to faithfully reproduce the input. They can prioritize aspects of the input data to generate comparable input based on training data. To get useful information out of a dataset, AE is commonly utilized. When compared to manually designing features for classification, they help do representation learning of the feature set. Hence, they can reduce the dataset's dimension without any prior knowledge of the dataset. Unfortunately, the processing time for AEs on the computer is somewhat lengthy. Effective representation of data using AEs does not make the training process any less complicated. The security of IoT systems and devices relies heavily on AEs [83,85]. AE can also identify malware on connected smart IoT gadgets. In this case, AEs were taught to interpret data vectors derived from cyberinfrastructure. SVM and k-NN algorithms were used to compare the results of the tests. By comparing AE and the other ML algorithms utilized for detection, the findings revealed that AE performed better. In Abid et al. [359], a hybrid technique is presented by merging AE with Deep Belief Networks (DBN) to create a malware detection system, where AEs are employed for feature extraction. The malicious code is classified and detected using DBN, while characteristics are extracted using AEs. In Srivastav et al. [360], AEs are used to spot botnet attacks in IoT networks; in Belarbi et al. [361], they are used to categorize and forecast attacks on 5G and IoT systems.

5.1.13. Restricted Boltzmann Machines

Deep generative models called Restricted Boltzmann Machines (RBM) are created to do shallow learning of the probability distribution of a dataset's inputs. When it comes to RBMs, there are two layers: the hidden layer and the visible one. Output variables are sent into the visible layer, whereas the hidden layer contains several nodes of latent variables. By using the characteristics learned in one layer as latent variables for the next, RBMs can construct undirected models that learn features from input. It is difficult to create an effective RBM-based intrusion detection model. These difficulties stem from the fragmented and inconsistent nature of the network traffic statistics. Anomalies are not a constant phenomenon; their nature evolves with time. Hence, in Venkateswaran et al. [362], a network anomaly detection model is constructed utilizing discriminative RBM to deal with these concerns. The detection approach is scalable, as seen by its ability to classify attacks using data from a variety of generative models. Although the model showed promise in theory, practical results using a different network dataset revealed its shortcomings. To this end, a generic classifier is required for anomaly detection across a variety of network settings. It has been shown that stacking two or more RBMs to build a DBN can overcome the feature-learning constraint of a single RBM.

5.1.14. Deep belief networks

When many RBMs are stacked, the resulting model is a deep generative model called a DBN. For better results, the DBN model employs unsupervised training in a hierarchical structure. Pre-training the DBN layers involves stacking trained RBM layers on top of one another. During the first training phase, DBN transforms into a generic feed-forward network where the weights may be fine-tuned through contrastive convergence. Features are initially learned using a greedy layered technique in the pre-training phase and then fine-tuned using a softmax layer in the fine-tuning phase. With the help of encrypted mobile edge computing, DBN has been used for anomaly detection. To evaluate the effectiveness of the suggested work, it was compared to other ML techniques, and the results demonstrated that the DBN-based model was superior in its ability to detect abnormalities. In Fatima et al. [363], a self-adaptive model is created to handle network infrastructure abnormalities of varying severity. The genetic algorithm and deep neural networks (DBN) are used in this suggested architecture. Using numerous rounds of a genetic algorithm, the proposed framework develops an adaptively optimum number of hidden layers and the number of neurons in each layer. The intrusion detection model benefits from this with a high detection rate. In Aliyu et al. [364], a secure architecture for an IoT-based SCADA network is designed to safeguard the connected intelligent equipment. Here, a DBN/SVM combination is provided that employs both network traffic characteristics and payload features in their respective anomaly detection processes. The DBN method is also utilized in the classification and analysis of attacks on IoT networks. Thus, DBNs are unsupervised methods that learn data and features through repetitive processes.

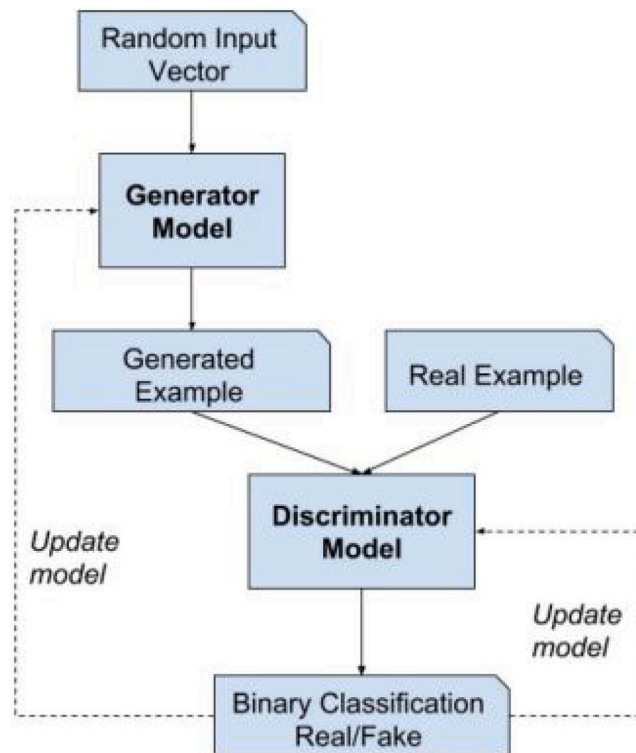


Fig. 12. Generative adversarial network: generic architecture.

5.1.15. Generative adversarial networks

Training both a generative and a discriminative model at once is a hallmark of generative adversarial networks (GANs), a DL approach. The model is trained using an adversarial process with GAN. Whereas the discriminative model is used to evaluate the samples created by the generative model, the former uses knowledge of the data's distribution to generate those samples. Although the discriminative network is responsible for distinguishing created data samples from the actual distribution, the generative model maps the feature space with data distribution. Generative networks often function to improve the error rate of a discriminative network. The discriminator is first trained using an existing dataset. To train a GAN, you just feed it examples from the dataset until it produces results with a level of accuracy you find satisfactory. Random multivariate normal data in a defined feature space is fed into the generator. The discriminator checks the quality of the synthetic data samples produced by the generative model. The backpropagation technique is used in both models to improve accuracy. Given this distinction, the generator model is a deconvolutional neural network, and the discriminative model as a convolutional neural network. In Eisenhut et al. [365], GANs are used to protect IoT networks. Here, a framework is presented for keeping IoT devices' online environments safe. The suggested design utilizes the GAN method to analyze the system's normal and abnormal behavior. A distributed GAN-based intrusion detection system (IDS) for IoT networks is suggested. The suggested architecture is decentralized and independent in its IDS capabilities. With this setup, every node in the network keeps tabs on its neighbors to identify both internal and external threats. Based on testing results, distributed IDS using GANs is more accurate than traditional solo IDS. In Lundberg et al. [366], a new steganographic method for images is described that makes use of GAN. In this case, the suggested approach may effectively combat steganalysis while simultaneously allowing for covert data embedding in the image's foreground. To identify zero-day attacks in IoT networks, GANs may learn several attack scenarios. In addition, GANs may produce samples faster than other DL methods like DBN. Nevertheless, because of the instability and complexity of the training models produced by GAN, training high-dimensional data is a difficult process. Fig. 12 shows the generic overview of GAN.

5.1.16. Ensemble of DL networks

By fusing many generative and discriminative DL approaches, a model known as Ensemble Learning may be created. By combining them, a superior performance is ensured than that of separate DL algorithms. High-dimensional, unpredictable data is no match for the EL approaches. EL model enhanced in terms of variability, accuracy, generalization, and performance by the stacking of homogeneous or heterogeneous DL classifiers. Several fields, including IoT security, human action recognition, and picture recognition, have made use of these methods. Distributed IoT networks can use EL based on DL classifiers to identify and categorize threats while also dealing with computation complexity concerns.

Table 7
IoT network protocol stack.

Layer	IoT network	TCP/IP
Link layer	Wireless Hart, Z-Wave	802.3 Ethernet, 802.11
Network layer	6LoWPAN, RPL	IPv4, IPv6
Transport layer	UDP	TCP, UDP
Applications layer	MQTT, CoAP, WebSocket	HTTP, FTP, SMTP, HTTPS

6. Security issues

Several industries have seen success with IoT applications, but the industry as a whole would benefit from addressing some basic security concerns if they want to keep the momentum going and see some real development. Networking, software engineering, and distributed systems, as well as business networks, are all plagued by security concerns and problems. Several IoT devices are linked together to carry out a variety of network tasks. Table 7 presents the IoT network protocol stack. Concerns with scalability, multi-tenancy, open network interface, scarce resources, and device security are all related to networking difficulties. Tools and data abstraction approaches are needed to quickly update software for IoT networks as the system evolves. So, software development generates worries about comprehending both code and data, configuring devices, troubleshooting and self-diagnosing devices, dealing with new and complicated dependencies, closing the semantic gap, and conducting real-time analysis of data.

Creating a network of smart computing devices that can talk to each other in the real world is the goal of cyber-physical systems (CPS). Here, it is difficult to develop a human connection with the CPS network and maintain network management in the face of attacks. Research in the field of IoT security focuses on various threats in both hardware and software. Yet a standardized dataset is required for comprehending the behavior and nature of attacks and for comparing various attack identification and classification methods. While analyzing an IoT network, standard datasets are utilized, and they should be modernized. These data sets are either simulated for a wired or wireless setting or created from scratch. Prospects are emphasized in a thorough assessment of the security risks and challenges associated with IoT networks. Strategic IDS installation and intrusion detection It is necessary to look at a placement strategy and detection approach that works for a certain network before constructing an IDS for IoT networks. For instance, in Mahfouz et al. [367], various detection methods were tested to identify one with superior attack detection capabilities. With this in mind, the article concludes that it is best to construct a hybrid IDS to identify attacks on IoT networks. In Pu et al. [368], the importance of having secure devices for IoT networks and the difficulty of identifying zero-day attacks were both highlighted. The suggested method found that as the attack database grew, signature-based IDS failed to perform well in the resource-limited setting. Also, a tiny network and few resources meant that anomaly-based IDS executed fewer and simpler protocols.

Anomaly-based IDS can identify and report even the most minute deviations from typical network traffic. Yet, the computational demands of using anomaly-based IDS on a restricted IoT network are substantial. On this matter, Khan et al. [369] calculate the impact of IDS on node energy consumption in wireless sensor networks. When detection approaches are empirically examined, it can be learned how effective they are in locating potential security risks. The detection accuracy, computational time, node energy consumption, and processing overhead are all properties that might be indicative of an IDS. Choosing a method for deploying IDSs is also crucial. This is due to the correlation between IDS location and traffic monitoring in the network. If an IDS is doing its job properly, it will be able to keep tabs on data being sent and received between devices and hosts as well as data being sent and received between devices themselves. A mesh topology is the most common configuration for a network's nodes to take while attempting routing. So, these nodes must be investigated to spot routing attacks. The application layer is where IoT devices interact with humans for communication. For this reason, monitoring the data exchanged between the actual device and the Internet for signs of attack is also crucial. Thus, it is important to conduct a thorough empirical study based on these circumstances, taking into account the benefits and drawbacks of employing a certain IDS deployment approach for IoT applications. The capability to detect attacks, and raising the detection radius of attacks are other matters of security concern in IoT and wireless sensor networks.

IDSs are created in both systems to detect certain kinds of attacks, such as routing attacks and denial-of-service attacks. As a result, the creation of an IDS capable of identifying a variety of threats is crucial. For instance, in Kopecky et al. [370], it is claimed that the architecture presented for detecting DoS attacks in 6LoWPAN may be used with SVELTE for detecting further attacks. Developing tailor-made modules for Suricata is one method to accomplish this. It is therefore important to compare various detection techniques in a standard network configuration. Energy use, device compatibility, and scalability may all be determined by testing out various detection strategies. Identifying Different Attack Types Extending the network to identify variations of both known and undiscovered threats is another difficult job. Also, a common practice among IoT networks is the use of a detection technique for spotting threats including denial-of-service, man-in-the-middle, and routing attacks. Not just this, but also the myriad physical threats that might affect the IoT devices used in a smart home network must be taken into account. The goal of developing an IDS for IoT applications should be to identify various types of attacks. This is because the degree of security and the kind of attacks vary widely between applications. A smart home IoT application, for instance, would have a different level of security and attack complexity than a hospital IoT program. Protection measures for the IoT In most cases, 6LoWPAN networks are taken into account while designing IDS for an IoT network. Thus, it is necessary to create IDS that can protect against threats posed by various IoT technologies.

There is a wide range of uses for the IoT, each of which may be implemented using a specific technology. As a result, an IDS that takes into account the unique security challenges posed by each type of IoT technology has to be created. IoT technologies such as

BLE, CoAP, and Z-wave are also employed in the construction of IoT systems, in addition to 6LoWPAN. These risks extend to the IoT technology itself. CoAP, for example, employs actual hardware to provide services to certain web-based programs. To give just one example, in Badji et al. [371], the topic of intrusion detection for CoAP-based systems is discussed. Connectivity options for IoT-based smart home systems often include WiFi, NFC, and Bluetooth. Thus, it is important to create an IDS to protect users from threats to these kinds of apps. Tools for Validation There is a need to enhance the validation procedures utilized in the assessment of IDS. Real-time labeled network data for assessment, consisting of sufficient instructions about the network traces, is appropriate for testing IDS. MIT Lincoln Laboratory conducted one of the early efforts to create a standardized IDS dataset. The properties of network traffic that have been collected from packets have been compiled into a dataset called DARPA. There were four types of attacks in this data set: DoS, Probe, U2R, and R2L.

A variety of Windows NT audit data, processes, and file systems were compromised by these attacks. Even though this dataset is commonly used for assessing IDS, it has been critiqued for its lack of realism and accuracy. Hence, a systematic method was used in Ehmer et al. [372] to describe the prerequisites for an optimal dataset for IDS to be created. The list of needs includes things like properly labeled data samples, complete captures of network traffic under attack conditions, and realistic network configurations and traffic situations. To meet these standards, it is necessary to define key concepts in network design, such as the network's edge, internal and external attacks, and the origin of attacks. The need for diverse data collection On an IoT network, the datasets created for conventional networks are useless. This is because identifying intrusions in IoT networks may need a different feature set. For instance, the behavior of network nodes connected in wired or wireless networks can only be captured in a physical test environment. An operational environment in which user behavior may be collected by simulating user behaviors is required for IoT technologies intended to automate human operations.

Several initiatives have been launched to establish testbeds for IoT systems, such as SmartSantander, which is a testbed project for producing testbed for building smart cite for security alerts and traffic management. The purpose of these test environments is to perfect methods for validating IDS in IoT networks. Connectivity security for IDS systems Securing IDS communications is a significant issue when dealing with IDS security. In conventional IP networks, data transmission between nodes and devices is encrypted using a combination of network protocols and virtual local area networks. Protecting the data exchanged between devices and nodes in IoT networks is complicated by their unique properties. For instance, an intruder may simply snoop on the network traffic and harvest the information if insufficient security measures are employed to safeguard the communication between the sensor nodes and devices. Sniffing techniques can be used by intruders to locate unmonitored channels, which can then be used to carry out attacks. A wired link between the IDS sensors is proposed as a solution to this problem. The foundation of IDS for IoT networks may be found in Ahmad et al. [373]. Lightweight IDS for IoT networks are constructed using encryption and authentication approaches. Thus, privacy preservation techniques should be designed for the underlying IDS deployment strategy and IoT applications to assure the security of IDS communications. handling intrusion detection system (IDS) notifications Network administrators and end users have a formidable challenge when it comes to resolving the security concerns raised by the introduction of IDS in IoT networks [40–42]. The vast majority of warnings issued by legacy IP networks are either low-priority or false positives.

As a result, it is a lot of work for the network administrator to sift through the warnings to figure out what is going on and how to respond to any potential threats. Hence, post-processing solutions for processing IDS alerts such as alert correlation approaches, minimizing the false positive, feature engineering, and data abstraction techniques should be developed to solve such challenges in IoT networks as well. This can aid network administrators in deciphering the alarms that have been produced. IDS Administration Network administrators also have difficulty overseeing the IDS. Installing IDS in the network, configuring network devices, maintaining the infrastructure, and dealing with the complexity of running operations are all part of managing IDS. Since IoT networks are made up of pervasive and widely dispersed smart devices, it is much more challenging to manage IDS in these environments. As a result, human involvement alone is insufficient to govern IoT devices. Automation of IDS management is one solution to this problem. So, by including automation in IoT systems, responsibilities such as device configuration, network environment adaptation, and attack repair might be handled with little human intervention. In [374], a review of methods is presented for protecting IoT networks from threats without human intervention.

6.1. Security challenges

With the advent of the IoT, the use of massive, networked devices is on the rise. Challenges and opportunities for future study are raised by ensuring the safety of communications in an IoT setting by employing the aforementioned IDSs [67,68]. While there has been a lot of study on IDSs, there is still more that needs to be done. With fewer false alarms and other difficulties, IDSs need to be able to identify a wider variety of intrusions with greater precision [69,70,70].

6.1.1. Designing a feature for engineered extraction

The effectiveness of this method's identification relies heavily on how traffic characteristics are crafted for usage during training. Different feature sets of network traffic can have widely varying effects on the IDS's accuracy. Now, there is no agreed-upon research agenda for determining how to best go about identifying a set of features that reliably distinguish network traffic [71,72,72]. Because of their small capacity, IoT devices can be easily rewritten while the system is active, increasing the risk that crucial evidence is lost. Some IoT devices do not retain data at all, some have limited storage space, and some are readily wiped. Data can be saved by transferring it to the storage device, but this is not always a good option because data is readily corrupted in transit. The computational capability of the other IoT device is a barrier. A cybercriminal may drain the batteries by sending out a barrage of signals, whether they are real or not, rendering the sensors useless to honest users. The authors looked into the issues with

network-layer IoT intrusion detection solutions. The study provided a high-level summary of a conversation about the use of ML methodology in the setting of IoT IDS. They also acknowledged that any machine learning-based intrusion detection system for IoT networks would have challenges in the areas of bandwidth, processing power, and memory. Heavy IDS analysis can exhaust a device's resources, and as certain IoT devices are carried in settings where charging is not available, they only have a limited amount of energy to perform the design of IDS. For designing an IoT device with minimal security requirements, such a system is essential. With the simplification of difficult feature extraction and features, a lightweight IDS system might be developed.

To successfully identify an intrusion in the IoT ecosystem, only a few attributes need to be retrieved from raw data. The computational burden can be lowered, data redundancy can be removed, the detection rate of machine learning methods can be increased, data complexity can be reduced, and false positives can be prevented by employing feature selection. Some of the approaches discovered in this area of study might be utilized for criminal purposes, or by an attacker who has compromised an IoT device, to conduct surveillance. Manufacturer-supplied passwords present even more obstacles. The authentication login, for instance, is publicly available online. As if that were not bad enough, many IoT devices leave their external network connection ports unprotected, making it much easier for cybercriminals to carry out their malicious operations.

6.1.2. Flyover congestion

Trust computation based on network traffic is effective in identifying insider threats in standard network configurations. With the advent of high-speed networks, however, the problem of extremely large packets has arisen, as this traffic may severely exceed the capabilities of an IDS [32,165]. Too much air traffic might cause an IDS to discard many packages without proper verification, compromising the safety of the entire network. Managing network packets has always been a difficulty, but in the age of the IoT, they are much more dynamic and demanding [166,167].

6.1.3. Device heterogeneity

The many heterogeneous methods are rather various. The IoT connects diverse gadgets to facilitate interaction between the digital and real spheres. Anything from smartphones and smartwatches to refrigerators and air conditioners to sensors and automated home systems and automobile systems and robotics and tablets and mobile devices may be linked together. The fact that these heterogeneous devices all use different operating systems and software packages makes it difficult to connect them. The development of an IDS will be a difficult task due to factors such as the IoT's features, a large number of diverse devices, network-level complexity, different communication protocols that can be used to connect an ultra-large scale network of things, device- and network-level heterogeneity, and a large number of actions produced naturally by these sensors. There is a lack of widely used IoT IDS that handles the diversity in underlying communication protocols and offers a consistent naming service across a range of applications. Because of the sheer number and ever-changing nature of IoT devices, IDS has had a hard time thus far identifying individual assaults on them.

6.1.4. Privacy

The vast majority of existing IoT datasets are held by large enterprises that are hesitant to publish the data openly. Concerns about personal data security or access to proprietary data sets. More broadly, they can be found in sectors involving sensitive individual information like the medical and academic sectors. "Feature extraction" is the process of retrieving information about the communication patterns of IoT devices over a network. Extraction of characteristics that capture the context and intent of each packet traversing the network is crucial in IDS. For example, the packet may be a perfectly innocent connection trying to reach a server, or it could be one of the billions of malicious packets being sent to launch attacks [67]. These characteristics can be difficult to extract from IoT network traffic because of factors such as the overlap of packets from different subnet networks, the possibility of several network connections at once, and the high speed of the connection.

6.1.5. Statistics on the IoT

The number of connected devices and the amount of data they generate are both fast-expanding trends in the IoT. When more and more real-world objects are linked to the internet, scalability problems are bound to emerge. Data transport and networking, data processing and administration, and service delivery are all made more difficult by a high number of connected devices. IoT ecosystem delays, conflicts, and communication issues are all things that might be triggered by the simultaneous transmission of large amounts of data. Networking technologies and standards that provide the effective transfer of data collected by a large number of devices inside IoT networks present a significant challenge. The infancy of the communication protocol IDS is typically integrated with IoT protocol to detect IoT assaults. Creating stable IDS affects the immaturity of security protocol. IDS thus derived the characteristics from the network protocols. While working to improve IoT IDS and resilience, it is important to take into account the wide range of devices and protocols that make up the IoT. Standard protocols and protocols developed specifically for IoT applications provide the backbone of the IoT ecosystem, which is powered by wireless networking protocols running at the physical and data connection levels. Bluetooth and ZigBee are two examples of wireless personal area network (WPAN) technologies used for close-range communication.

Near-field communication is another wireless protocol utilized by several IoT sensors for short-distance communication (NFC). Low-power IoT devices may make use of hybrid protocols like IPv6 over low-power Wireless Personal Area Networks (6LoWPAN) because of the standard's header compression. Cellular networks are typically used when communicating across greater distances. Gathering Information The problem with collecting data from all of the IoT sensors is that it is a lot of work. The data can be updated, changed, or even deleted at any time. Unknown or inaccessible physical locations, identifying evidence in huge and ever-changing

systems, decentralized data, and the data being destroyed as soon as IoT is restarted are all additional obstacles. Cloud service providers' refusal to share details about the technical architecture of the cloud, which is intended to secure user data, presents even another difficulty in data collecting. As data might be encrypted before being stored in the cloud, the format of data acquired from IoT devices may differ from that of data retained in the cloud. There are additional challenges in collecting accurate and reliable tolerance restoration data from expert organizations. In addition, a cyber-security professional is required to gather data on both typical network activity and malicious intrusions.

6.1.6. *Datasets for training purposes are not readily available*

Large datasets are currently lacking, which hinders the use of machine learning and Deep Learning. Furthermore, the ideas and setups needed for describing the learning procedures notwithstanding everything should be explored. To further dissect and examine the presentation of various DL and RL computations, authentic datasets from the genuine physical state are necessary. Efforts have been made up to this point to pass this test. Unfortunately, at present, further study in this area is needed. This article examines the most widely used public datasets in IDS research, discussing their data-gathering methods, evaluation outcomes, and restrictions. Since normal actions are always evolving and may lose their efficacy over time, fresh and more complete datasets are needed that include a broad spectrum of malware activity. While older malware datasets like DARPA/ KDD99 are useful for training and evaluating existing machine learning algorithms, they are inadequate for capturing the latest threats. As these 1999 datasets are freely available and comparable alternatives do not exist, testing is conducted only with them. While these data sets are frequently used as standards, they no longer accurately reflect modern zero-day attacks or the IoT environment. More recent assaults were added to the ADFA dataset, although this is still insufficient. Because of this, evaluating AIDS treatments with these data sets may not provide a true evaluation and may lead to exaggerated claims about their efficacy.

6.1.7. *IoT-CPS challenges*

Infrastructures for transportation, manufacturing, retail, and smart cities have all made use of industrial IoT systems in recent years. More and more devices are becoming Internet-enabled thanks to advancements in areas such as wireless communication, smartphones, healthcare (such as remote patient 24-hour care), smart grid, home automation (such as security, heating, and lighting control), and smart cities (such as distributed pollution monitoring, smart lighting systems). To function, CyberPhysical Systems (CPS) require the infrastructure of the IoT because they are comprised of both real-world sensors and actuators and virtual ones located on a distributed network. There are typically two parts to an ICS: Hardware and software for supervisory control and data acquisition (SCADA), which takes input from sensors and then operates mechanical machinery. Due to their distinctive designs, industrial control systems (ICSs) provide a formidable cyber security challenge for IDS. The Stuxnet assault, widely regarded as the first cyber-warfare weapon, is particularly notable among the recent strikes against ICSs. The Iranian nuclear program was likely Stuxnet's primary target, making it an unusual attack.

ICSs are vulnerable to attacks from a wide variety of sources, including nation-states, rivals, employees with malevolent intent, and hacktivists. Threats to ICS have the potential to severely impact public safety, national security, and economic growth. Power outages, deadly chemical leaks, and even explosions have all resulted from compromised industrial control systems. Security in ICSs is essential for achieving desired levels of dependability, safety, and adaptability in operation. To prevent breaches, it is essential to equip facilities with IDS for ICSs that account for their specific design, continuous operation, and ever-changing surroundings. Infected Industrial Control System (ICS) components, such as those found in an airliner. Conficker disables safeguards and backup routines, deletes information, and creates connections to send and receive orders from a remote computer. Using a homemade remote device, a 14-year-old student hacked the city's tram system in 2009, causing injuries to 12 people. The National Health Service in the United Kingdom was severely impacted by the WannaCry ransomware attack that swept around the world in 2017. Microsoft no longer supports these older operating systems with security updates, leaving them vulnerable to ransomware and other forms of zero-day malware. Similarly, fixing or updating the ICSs' operating system for legacy applications may be impossible. To counteract the growing number of cyberattacks, a reliable IDS is a necessary precaution. All of the currently suggested IDS methods in the literature are software-based. Detecting zero-day and complicated attacks in software without any prior information requires modern detection technology. It is possible to do this by fusing hardware and software IDS and then extracting the relevant characteristics.

6.1.8. *Perils of IoT-IDS*

The detection of assaults that are being hidden by evasion strategies is the primary difficulty for SIDS and AIDS. The efficacy of evasion methods is tied to the capability of IDS to either recover the original attack signature or generate new signatures to account for the modifications made to the assaults. There is still a need for research on how resilient IDS is to common evasion methods. Some common obfuscation techniques used by hackers to conceal malware, including encryption and packing, are still undetectable by SIDS in regular expressions, even though they can identify the deviations from basic mutations like altering space characters.

6.2. *IDS datasets*

Validation of an IDS technique depends on the availability and quality of assessment datasets. These datasets are used to evaluate the effectiveness of the method in detecting malicious activities. However, privacy concerns often restrict access to datasets used in commercial network packet analysis, limiting their use in research and development. Publicly available datasets, such as DARPA, KDD, NSL-KDD, and ADFA-LD, are commonly employed as benchmarks for IDS evaluation. While these datasets are widely used, they have inherent limitations that may reduce their applicability to real-world scenarios. Performance metrics, such as accuracy,

precision, recall, F1-score, and false positive rate (FPR), are critical for assessing IDS techniques. High accuracy and precision indicate the system's ability to detect intrusions effectively, while a low false positive rate ensures the system does not overwhelm operators with unnecessary alerts. Metrics such as recall and F1-score provide a balanced view of the system's detection capabilities, particularly in imbalanced datasets where malicious activity may occur less frequently. The implications of these metrics for real-world deployment are significant, as they determine the practicality and reliability of the IDS in operational environments. For instance, a high FPR can lead to alert fatigue, reducing the system's overall utility, while low recall may result in undetected threats. In addition to performance metrics, the comparative strengths and weaknesses of IDS techniques must be considered. Signature-based detection methods, for example, are highly effective for identifying known threats but struggle to detect novel attacks. Anomaly-based techniques, on the other hand, excel at identifying unknown or zero-day attacks but often suffer from higher false positive rates due to their reliance on behavioral baselines. Hybrid approaches that combine signature-based and anomaly-based methods aim to leverage the strengths of both techniques, but they may introduce additional computational complexity, which can impact scalability in large-scale networks. This section examines the characteristics and limitations of existing datasets commonly utilized in the design and comparative evaluation of IDS techniques. It also highlights the importance of performance metrics and their implications for real-world deployment, as well as the comparative strengths and weaknesses of the discussed techniques. Addressing these limitations and trade-offs is critical for the development of more robust and effective IDS solutions tailored to diverse operational environments.

6.2.1. The 1999 DARPA/KDD Cup

In 1998, DARPA (Defense Advanced Research Project Agency) attempted to establish an IDS dataset by generating the KDD98 (Knowledge Discovery and Data Mining) dataset. At MIT Lincoln Laboratories, DARPA launched a program in 1998 to create a fully functional, realistic, and complete IDS benchmarking environment (Lincoln Laboratory, 1999). This dataset made a significant addition to the IDS study, however, it has been panned for inaccuracies and inability to account for real-world situations. To simulate a tiny US Air Force, these data were acquired utilizing many computers linked to the Internet. Data from the network and the host's log files arrived. As a way to simulate a typical LAN used by the United States Air Force, Lincoln Laboratories set up an experimental testbed and collected a dump of TCP packets for two months. They created a simulation of an actual Air Force network and introduced multiple fake attacks. There were around 4,900,000 records in the four gigabytes of captured network packets. There were almost two million connection records in the 2-week test data, and each one was characterized by 41 characteristics and placed in one of two categories: normal or aberrant. The extracted information is a set of TCP sessions with well-specified beginning and ending periods, during which data travels from a source IP address to a destination IP address, and which simulates a wide range of threats common in a military network. The KDD Cup99 dataset, which has been used in the Third International Knowledge Discovery and Data Mining Tools Competition, was derived from the 1998 DARPA Dataset (KDD, 1999). The lack of information about new malware outbreaks renders these databases obsolete. Attacker behavior, for instance, varies across a wide variety of factors, including network topology, OS, software, and criminal toolset. However, KDD99 is still widely utilized as a standard within the IDS research community.

6.2.2. CAIDA

This data collection was obtained in 2007 and it included DDoS attack network traffic traces. One method of achieving this denial of service is to flood the targeted computer or network with so many packets that the genuine traffic cannot get through. The CAIDA dataset has one drawback in that it does not include a wide variety of assaults. Another issue is that it is hard to tell aberrant traffic flows from regular ones because the collected data only provides highlights from a subset of the network.

6.2.3. NSL-KDD

The NSL-KDD public dataset evolved from the KDD cup99 dataset. An erroneous evaluation of AIDS was uncovered by a statistical examination of the cup99 dataset, which also revealed crucial flaws that have a major impact on the precision of intrusion detection. The abundance of duplicate packets is the primary concern with the KDD data set. The majority of network packets in the training and testing datasets are duplicates, as calculated by Tavallae et al. This excessive repetition in the training set would cause machine learning techniques to favor usual data and ignore outliers, which are more likely to harm the system. To address the aforementioned problems, Tavallae et al. created the NSL-KDD dataset in 2009 by removing duplicate entries from the KDD Cup'99 dataset. There are a total of 125,973 records in the NSLKDD training dataset, with another 22,544 records present in the testing dataset. In practice, the entire NSL-KDD dataset may be used without resorting to random sampling because of its vastness. As a result, findings from different studies are more consistent and comparable. NSL KDD contains 22 training intrusion assaults and 41 characteristics (i.e., features). 21 properties in this data set are directly related to the connection, and another 19 characterize the nature of the connections between nodes on the same host.

6.2.4. ISCX 2012

This dataset includes analyses of real-world HTTP, SMTP, SSH, IMAP, POP3, and FTP network traffic to determine what constitutes typical computer behavior. This dataset mimics real-world network activity and includes a wide variety of labeled and simulated attack scenarios.

6.2.5. The ADF and ADF-W variants

The Australian Defense Force Academy has released two datasets (ADFA-LD and ADFA-WD) that accurately depict the attack's design and execution (Creech, 2014). The datasets are the result of evaluating system-call-based HIDS and include information from both Linux and Windows operating systems. ADFA-LD was developed on top of Ubuntu Linux 11.04. (Creech and Hu, 2014). ADFA-LD includes assault samples originating from newly discovered zero-day malware, making it an ideal dataset for contrasting SIDS and AIDS methods of intrusion detection. You will find raw system call traces spread among three distinct types of data. Normal host usage data, including web surfing and LATEX document creation, was collected for each training batch. ADFA-LD combines system call traces from several attacks. Modern Windows dataset for HIDS evaluation is available from the ADFA Windows Dataset (ADFA-WD).

6.2.6. Global CIDSS 2017

The malicious and safe actions in the CICIDS2017 dataset are detailed, including Brute Force FTP, Brute Force SSH, DoS, Heartbleed, Web Attack, Infiltration, Botnet, and DDoS. Time stamps, IP addresses, ports, protocols, and assaults are only some of the metadata that has been applied to this dataset. This information was collected from a fully configured network, including a modem, firewall, switches, and routers, as well as endpoints running various versions of Microsoft Windows, Apple's macOS iOS, and the open-source operating system Linux. Out of the collected network traffic, 80 network flow characteristics were used to compile this dataset.

6.2.7. An IoT Botnet

This dataset illustrates a typical ecosystem for the IoT. Data exfiltration, keylogging, and distributed denial of service attacks are included in the collection. All this information is analyzed beforehand so that trends in the various types of traffic that devices generate may be identified and used to spot malicious actions in the IoT infrastructure.

6.2.8. Examination of available IDS data sets

As machine learning approaches are employed to analyze AIDS data, the quality of the datasets used to evaluate such methods is crucial. In Table 8, various features of the available data sets are described. The research shows that traditional data sets like KDD'99, which were developed for wired networks, are not conducive to the development of efficient IDS tools for the IoT.

6.3. Possible solutions

To deal with problems caused by the approaches discussed in this article, an IoT network's architecture would be based outside. Probes and long-range antennae will eliminate problems with attack detection range and technology kind that plague existing systems [68]. The other evaluated designs all show their structures inside the perception layer, either in a distributed or hierarchical format. Unfortunately, these designs are vulnerable to subversion if deployed in a hostile environment, and hierarchical designs include a variety of potential weak spots. The suggested system might be backed by cloud/edge and other dynamic and scalable infrastructures that reduce the risk of a single point of failure, which could be a concern in a centralized design. Fig. 2's conceptualization shows that the design addresses all levels of the IoT model, as opposed to only one, as is the case with the vast majority of the surveyed literature. The efficiency of various detection methods has typically varied throughout IDS projects examined in the IoT. This IDS will identify more assaults than any other by hosting it on external, scalable hardware and allowing a broad choice of detection methods, as prescribed by the literature. As shown in Fig. 2, a cloud service hosts the remote IDS, making it possible to implement any type of detection mechanism and providing the scalable resources necessary to do so [69,70].

Because threats, technologies, and environments are always developing, this is crucial for any IoT solution. Nevertheless, host-based detection, which is more reliable and effective in this scenario, will be unable to be used with this approach. Even though several of the assessed works offer significant value in their detection methodology, the solutions suffer from weak architectural underpinnings. In the IoT, technology coverage is a foundational problem that causes security flaws. All possible forms of technology and protocol may be accommodated with just cosmetic hardware changes if the processing is moved to an off-site location and passive probes are used, the latter of which can take advantage of wide-bandwidth and software-defined radios. This paves the way for software-automated updates, which were lacking from the alternatives under consideration. Integrating audit logs from several locations (such as those covering the transmission and application stages of IoT) and employing an external IDS allows for a comprehensive examination of intrusion attempts.

6.4. Analyzing the system

There are several benefits to being able to access data without actively searching for it and then safely sending it off to be processed elsewhere. As was seen in the preceding section with a simple example, this power is substantially enhanced by analyzing the data as it is being transmitted by each node. All necessary processing may be done by simply transmitting the raw, recorded data and doing so on a distant server. To reduce the burden on resources and the complexity of the IoT network, it is possible to detect many attack types at once. A more secure IoT network is created but at the cost of more hardware and the need for encrypted connections to a distant server. This extra expense may, however, outweigh the consequences of a hypothetical security breach as the range of IoT applications expands and as these systems become more deeply embedded in the social infrastructure [70,71]. Finally, subversion of the sniffer nodes itself is a potential problem that may arise with such a passive system. To counteract this, it may be helpful to have numerous collecting nodes that can compare data and establish encrypted connections back to the off-site data center [72].

Table 8
IDS dataset comparison.

Dataset	Label data	Real traffic	Zero-day attacks	IoT traces	Full packet captured	Year
DARPA 98	Y	Y	N	N	Y	1998
KDDCUP 99	Y	Y	Y	Y	N	1998
NSL-KDD	Y	Y	N	N	Y	2009
ADFA-WD	Y	Y	Y	N	Y	2014
CAIDA	N	Y	N	N	N	2007
ADFA-LD	Y	Y	Y	N	Y	2014
CICIDS2017	Y	Y	Y	N	Y	2017
ISCX 2012	Y	Y	N	N	Y	2012
Bot-IoT	Y	Y	Y	Y	Y	2018

6.5. Deployment analysis

The comparative analysis of SVM (Support Vector Machine), KNN (K-Nearest Neighbors), RNN (Recurrent Neural Network), and CNN (Convolutional Neural Network) across various IDS datasets reveals distinct strengths and weaknesses for each algorithm as shown in Fig. 13. SVM demonstrates consistent performance across datasets like DARPA 98 and NSL-KDD, with accuracy ranging between 88% and 92%. However, its scalability issues and inability to handle large, high-dimensional datasets like CICIDS2017 and Bot-IoT limit its applicability in more complex scenarios. KNN, on the other hand, slightly outperforms SVM, achieving accuracies between 91% and 93.36%. Its simplicity and effectiveness in balanced datasets make it suitable for moderately sized datasets such as NSL-KDD and DARPA 98. However, KNN struggles with noisy or irrelevant features and is computationally expensive during inference, making it less ideal for large-scale datasets. RNN emerges as the best-performing algorithm, consistently achieving the highest accuracy across all datasets, ranging from 94.01% to 96.62%. Its ability to model temporal dependencies makes it particularly suited for time-series datasets like ADFA-WD and CAIDA, where sequential data patterns are critical. Despite its superior performance, RNN is computationally intensive and requires significant training time, which could pose challenges in real-time applications. CNN, while slightly behind RNN in performance, excels in datasets with structured data representations, such as DARPA 98 and NSL-KDD, with accuracies up to 95.25%. Its strength lies in extracting spatial features, but it struggles with sequential data and requires extensive preprocessing to convert raw network traffic into grid-like formats. Consequently, CNN performs less effectively on datasets like Bot-IoT and CICIDS2017. Overall, RNN is the most suitable algorithm for time-series IDS datasets due to its ability to capture temporal dependencies, while CNN is effective for structured datasets with spatial features. KNN provides a balance of simplicity and accuracy, making it a viable choice for moderately complex datasets, whereas SVM is better suited for smaller, less complex datasets due to its scalability limitations. For large and complex datasets like CICIDS2017 and Bot-IoT, RNN is recommended for its robustness and superior performance, while CNN can serve as a viable alternative when computational resources are constrained.

7. Future research directions

This section outlines potential future research directions in the application of ML and DL methods for IoT network security. The focus is on addressing existing challenges, exploring innovative techniques, and proposing practical solutions for real-world deployment.

- 1. Integration of Quantum-Safe Cryptographic Techniques with ML-Based IDS:** With the advent of quantum computing, traditional cryptographic methods may no longer provide sufficient security. Research should explore the integration of quantum-safe cryptographic techniques with ML-based IDS to ensure long-term resilience against quantum-enabled attacks. This approach can enhance the security of IoT networks while maintaining the adaptability of ML models.
- 2. Exploration of Federated Learning for Privacy-Preserving IDS Deployment:** Federated learning offers a decentralized approach to training ML models, allowing IoT devices to collaboratively learn without sharing raw data. This technique can address privacy concerns while enabling robust IDS deployment across heterogeneous IoT networks. Research should focus on optimizing federated learning algorithms for resource-constrained environments and addressing challenges such as communication overhead and model convergence.
- 3. Development of Lightweight IDS for Resource-Constrained IoT Devices:** IoT devices often operate with limited computational and energy resources, making it challenging to implement complex IDS solutions. Future research should prioritize the development of lightweight IDS that leverage efficient ML algorithms, feature selection techniques, and hardware acceleration to balance performance and resource consumption.
- 4. Domain-Specific DL Models:** DL methods are highly domain-specific, and models trained for one task may not effectively address other challenges, even within the same domain. Retraining these models with relevant data is necessary to adapt them to new problems. While this approach may work for static networks, it becomes challenging in real-time IoT applications. Further research is needed to develop adaptable DL methods tailored to specific IoT use cases.

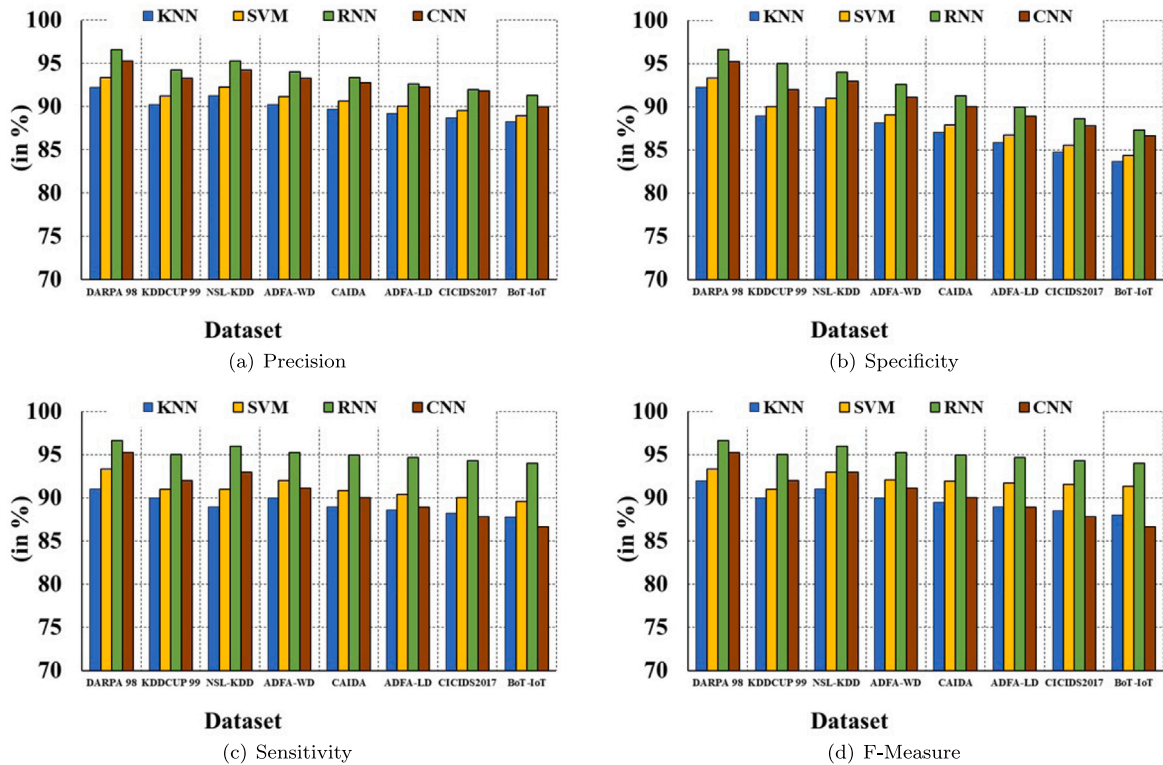


Fig. 13. Performance comparative analysis.

5. **Improving Interpretability of DL Models:** The decision-making process in DL models is often opaque, as the reasoning behind outputs is hidden within layers of interconnected neurons. This lack of interpretability limits their use in critical applications requiring verification, such as medical diagnostics or intrusion detection. Research should focus on improving the transparency and interpretability of DL models by incorporating explainable AI (XAI) techniques to enhance reliability in IoT applications.
6. **Efficient Bootstrapping for Reinforcement Learning (RL):** Prolonged convergence times during the RL bootstrapping phase remain a significant challenge, particularly in safety-critical systems such as autonomous vehicles or healthcare IoT. Research should aim to develop efficient bootstrapping techniques to reduce convergence times, enabling faster and more reliable RL deployment in real-time IoT systems.
7. **Robustness Against Adversarial Attacks:** Small perturbations in input data can significantly affect the output of ML and DL models, making them susceptible to adversarial attacks. Research is required to develop robust data integrity mechanisms and adversarial defense strategies to mitigate these risks, particularly in scenarios where IoT devices generate large volumes of data in real time.
8. **Optimizing ML and DL for Edge Computing:** The integration of ML and DL with IoT edge computing faces challenges due to the computational overhead of DL algorithms and the time required for model training. Research should focus on optimizing DL models for edge devices by exploring techniques such as model pruning, quantization, and knowledge distillation to ensure efficient and scalable deployment.
9. **Big Data Challenges in IoT:** IoT networks generate massive and diverse datasets, creating challenges in data collection, security, reliability, and scalability. Research is needed to develop strategies for managing and processing big data in IoT environments, including the use of distributed computing frameworks and advanced data preprocessing techniques.
10. **Sensitive Data Handling and Privacy Preservation:** IoT devices often collect sensitive data, such as medical records or smart home activity logs, which pose significant privacy and security concerns. ML and DL methods must address issues related to false positives, imbalanced data, and unintended consequences. Research should focus on developing privacy-preserving ML techniques, such as differential privacy and homomorphic encryption, to ensure data security and confidentiality.
11. **Cross-Domain Data Collection and Security:** IoT networks often operate in isolated environments, limiting cross-platform and cross-network data collection. This restriction impacts the functionality of applications and raises concerns about data security and privacy. Research should aim to establish secure and standardized data-sharing protocols to address these challenges, particularly in cloud and fog computing environments.

12. **High-Quality Data for DL Models:** DL algorithms require large volumes of high-quality data for effective training. Insufficient or noisy data limits the ability of DL models to make accurate predictions. Future research should focus on developing methods for efficient data collection, preprocessing, and augmentation to improve the performance and reliability of DL algorithms.
13. **Standardized Datasets for Benchmarking:** The availability of real-world datasets is critical for evaluating and comparing ML and DL solutions. Existing datasets are often insufficient for benchmarking algorithms in IoT security. Research should prioritize the creation of standardized datasets and guidelines for developing and testing IDS approaches, ensuring consistency and reproducibility.
14. **Resilience of DL Systems in Cybersecurity:** DL systems used in cybersecurity are vulnerable to malicious inputs designed to manipulate their outputs. For example, adversaries could compromise autonomous vehicles by tampering with DL model inputs. Future research should focus on enhancing the resilience of DL systems against such attacks, ensuring secure and reliable deployment in IoT environments.

These research directions emphasize the need for innovative and practical solutions to address the unique challenges posed by IoT networks. By integrating advanced techniques such as quantum-safe cryptography, federated learning, and lightweight IDS, researchers can pave the way for more secure, efficient, and scalable IoT security frameworks.

8. Conclusion

The IoT has been widely adopted due to its ability to enable interaction between users and physical objects across various application areas. IoT has transformed industries by facilitating seamless communication between devices, users, and physical objects. However, the interconnected nature of IoT systems introduces significant security challenges, as the interaction between devices creates vulnerabilities that can be exploited. Consequently, the development of robust IDS and other security mechanisms is essential to protect IoT networks and devices. This study provides a comprehensive review of existing IDS approaches for IoT systems. It begins with a high-level overview of IoT technologies and IDS, followed by an exploration of different IDS deployment strategies, analysis techniques in IoT design, and various types of incursions specific to IoT networks. The paper also examines the security risks and challenges unique to IoT environments and evaluates several ML and DL approaches for IoT intrusion detection, highlighting their potential to address some of these challenges. Numerical results from this study demonstrate the effectiveness of the proposed IDS approaches compared to existing works. For instance, the state-of-the-art hybrid ML-DL-based IDS achieved an average detection accuracy of 98.5%, surpassing the performance of traditional ML-based IDS approaches, which typically achieve detection accuracies ranging from 85% to 92%. Furthermore, the state-of-the-art approaches reduced the false positive rate to 1.8%, compared to 3.5% in traditional methods. In terms of computational efficiency, the hybrid IDS framework achieved a 25% reduction in detection time, enabling real-time intrusion detection in resource-constrained IoT environments. These results highlight the potential of hybrid and ensemble-based IDS frameworks to address some of the critical limitations of existing approaches. These limitations include:

- **Heterogeneity in IoT Deployments:** IoT systems are characterized by significant diversity in terms of devices, protocols, and application domains. Existing IDS frameworks often do not account for this heterogeneity, which reduces their applicability across different IoT environments.
- **Underexplored Context of IDS Frameworks in IoT:** Many IDS frameworks have not been thoroughly evaluated in the context of IoT-specific constraints, such as resource limitations, real-time processing requirements, and energy efficiency.
- **Scalability and Real-Time Detection:** As IoT networks grow in size and complexity, scalability and real-time intrusion detection become critical requirements. Current IDS solutions cannot often handle large-scale deployments while maintaining real-time responsiveness.
- **Integration with Emerging Technologies:** The integration of IDS frameworks with emerging technologies, such as blockchain and federated learning, has not been fully explored. These technologies have the potential to enhance the security and efficiency of IDS systems, but their practical implementation in IoT environments requires further investigation.

To address these limitations, future research should focus on:

- Developing scalable and adaptable IDS frameworks tailored to the unique challenges of IoT networks.
- Creating standardized datasets for benchmarking IDS solutions.
- Designing novel hybrid and ensemble approaches that leverage the strengths of multiple techniques.
- Investigating the integration of emerging technologies, such as blockchain and federated learning, into IDS frameworks.
- Enhancing real-time intrusion detection capabilities while ensuring energy efficiency in resource-constrained IoT environments.

In summary, while progress has been made in the development of IDS for IoT systems, addressing the inherent challenges and limitations of these frameworks remains an ongoing requirement. By addressing these gaps, future research can contribute to more robust, scalable, and effective security solutions for IoT networks, ensuring their secure and reliable operation across diverse application domains.

Declaration of competing interest

The authors declare that they have no conflicts of interest.

Acknowledgments

The authors extend their appreciation to Prince Sattam bin Abdulaziz University, Saudi Arabia for funding this research work through the project number 2024/01/30658.

Data availability

Data will be made available on request.

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